رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

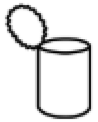
Analyzing Ecommerce data in Tableau

Session 4

Dataset

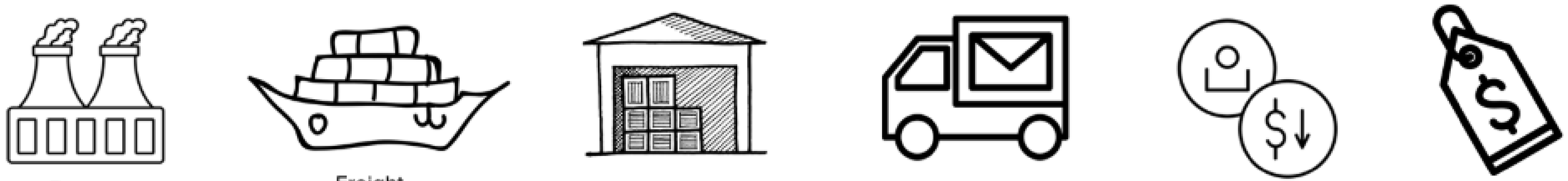
- The data is for an Online Store selling pet supplies.
- Please read metadata file for data description before start working
- Have a look on each data file

E-commerce operations and terminologies



Can pet food (4)

Illustrative total costs for canned pet food through various stages in the supply chain.



Factory		Freight		Warehouse		Shipping		Profit/loss		Price
$4	+	$1.5	+	$1.5	+	$5	+	8	=	$20
COGS (cost of goods sold)		Landed Cost (freight)		Fulfillment Fees		Shipping Cost		Profit (Loss)		Retail Price

Business Goals

- Increase Sales
- Reduce Expenses

- For sales growth, you will focus on upsell and cross-sell opportunities.
- In a cross-sell, you promote a relevant product at the point of purchase. For example, when a customer is ready to buy a beach umbrella, present an offer on beach chairs.
- In an upsell, you promote a higher-priced alternative or a higher quantity of a product. For example, when a customer buys pet food regularly, present an offer on organic food.

Business Goals

- Increase Sales
- Reduce Expenses

Shipping expense reduction strategies

For shipping cost reductions, we will explore the following alternatives. Consolidating multiple shipments into a single one. Shipping from a location closer to customers. Reducing package size dimensions and weight. And, finally, shipping a higher quantity of a product.

Business Goals

- Increase Sales
- Reduce Expenses

Market basket analysis

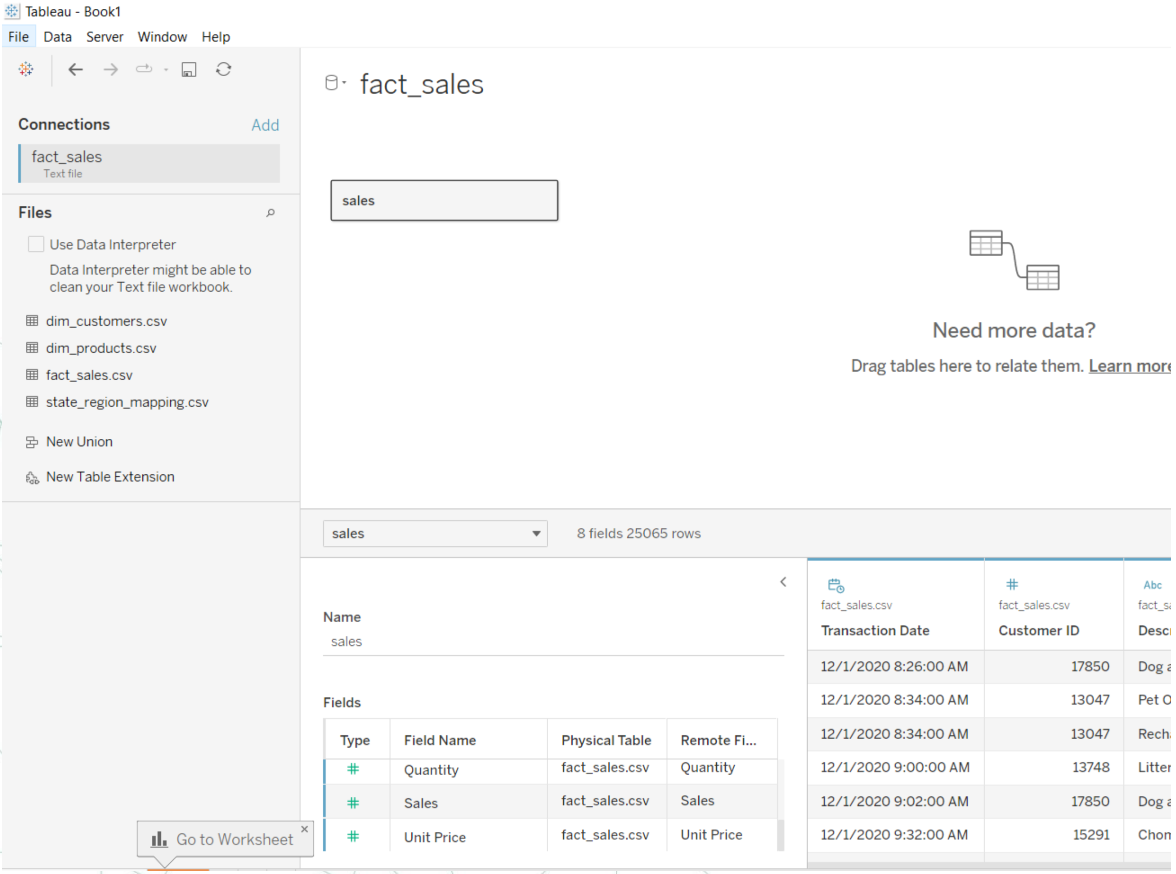
A market basket analysis aims to find products frequently bought together in a single checkout. This is determined by finding out the correlation coefficient between purchased quantities.

The value indicates the strength of the linear relationship. In this illustrated example, milk and bread correlate more strongly than milk and butter.

Let's have fun and start the real work

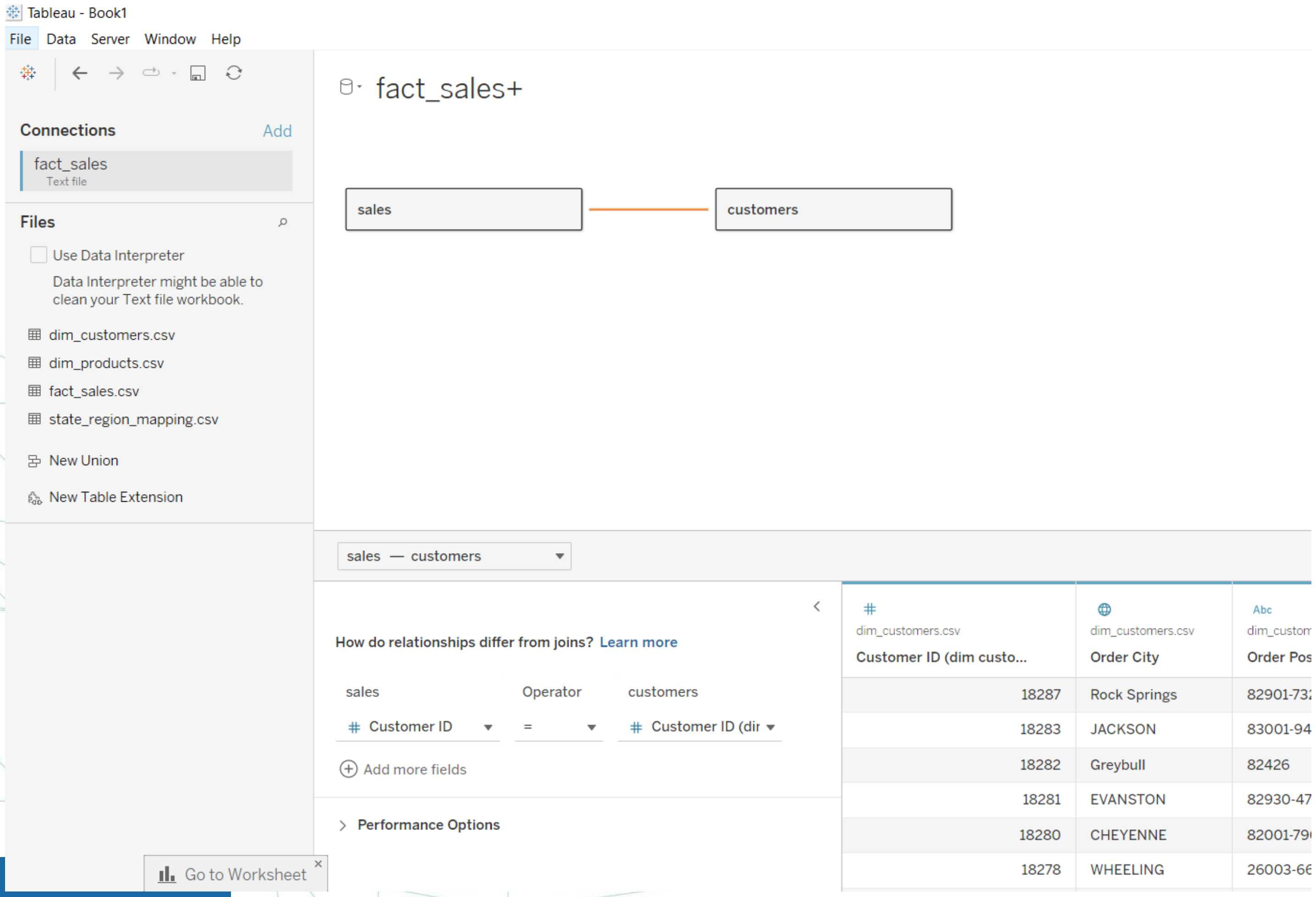
Step 1 – Data source connection

- **Connect the fact sales table and change its name to “sales”**



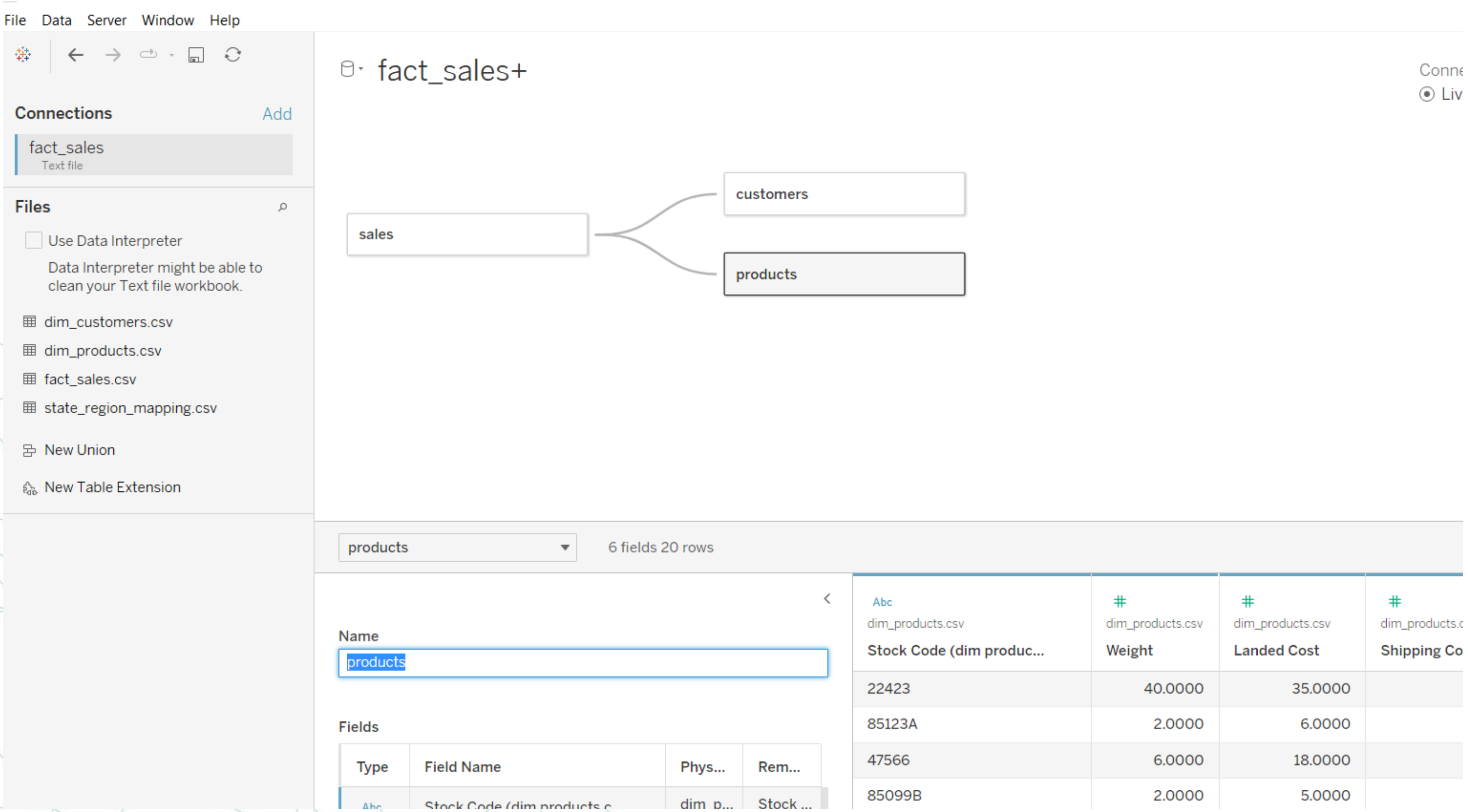
Step 2 – Data source connection

- **Drag the dim_customers table and change its name to “customers” and make relationship with the sales table**



Step 3 – Data source connection

- **Drag the dim_products table and change its name to “products” and make relationship with the sales table.**



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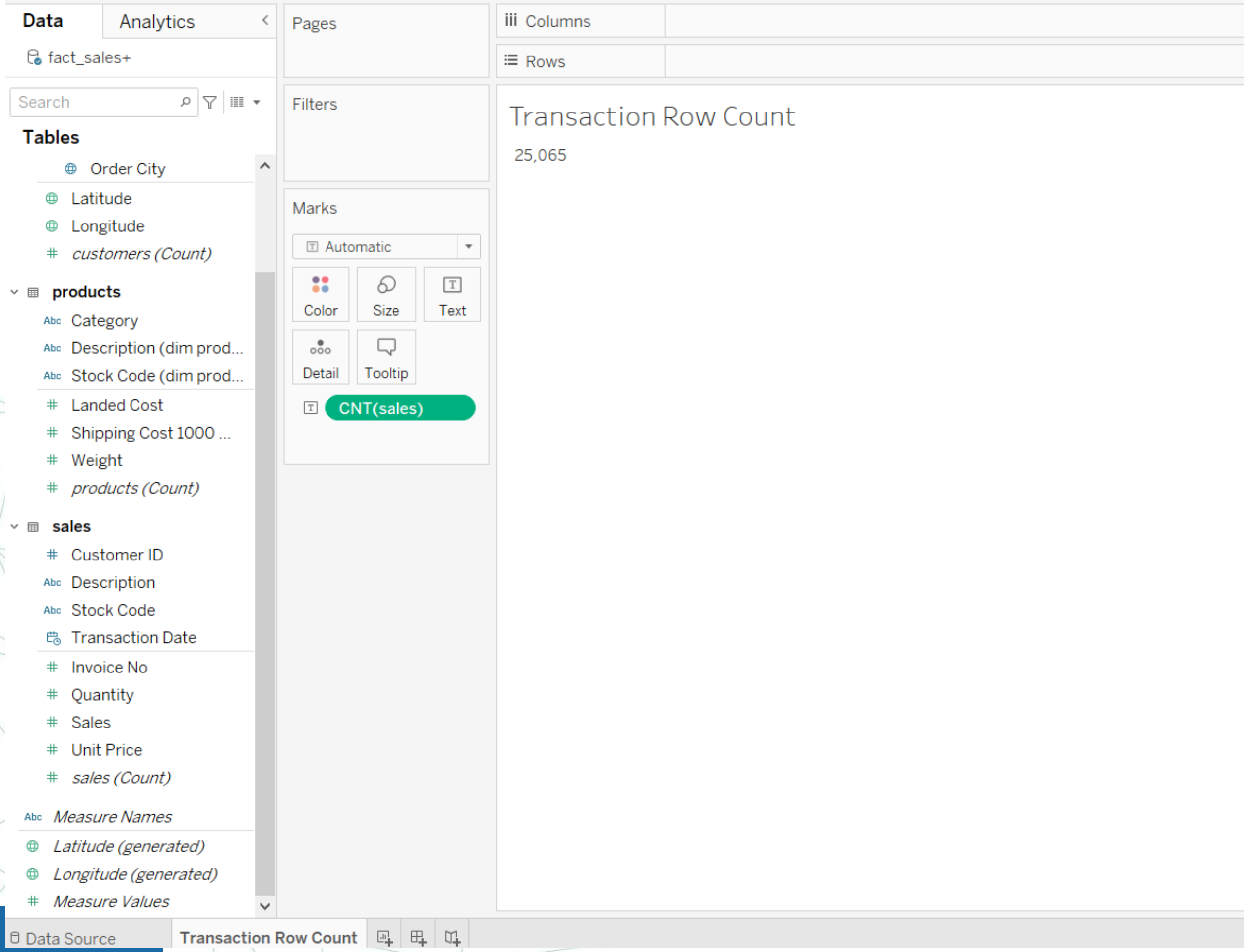
Step 4 – Row count

- Make your first worksheet name it “Transaction Row Count” and check the row count



Step 5 – Data Cleaning

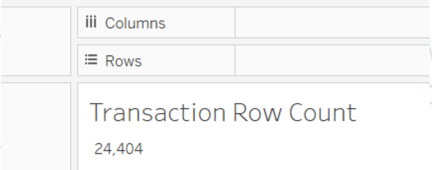
- All Valid Transactions must have a valid Invoice No.
- Create a new calculated field to flag the invoice no when it is null.

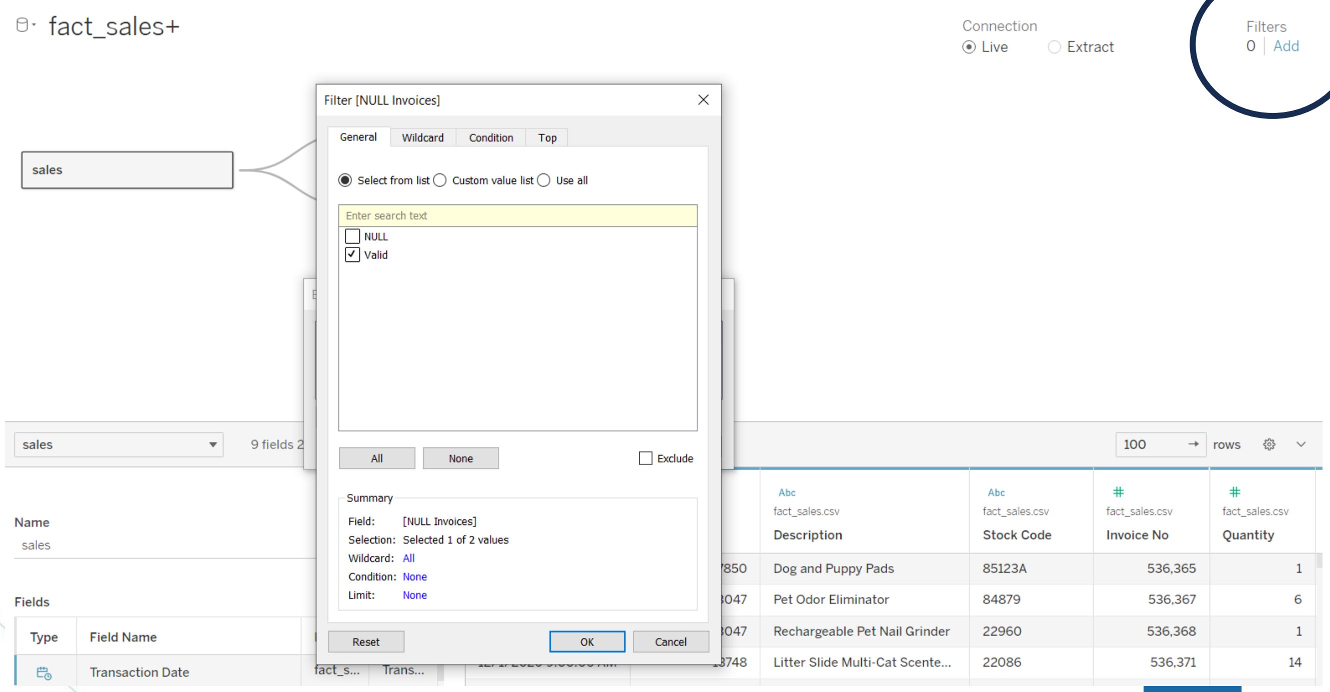
NULL Invoices

```
IF ISNULL([Invoice No]) THEN "NULL" ELSE "Valid" END
```

- Remove any records where the invoice no is null using data source filter.

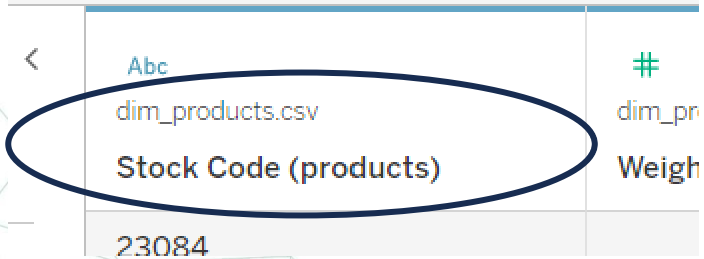
- How many records in the sales table after removing nulls? It should be 24404





Step 6 – Data Cleaning

- The data source has three tables that were renamed. The dimensions and measures are still suffixed with a table name.
- Rename each field with the correct table name suffix.
- For example, rename **Stock Code (data_products.csv)** to **Stock Code (products)**.



Step 7 – Building Metrics

- Now that you have a clean dataset to work with let us build a few metrics for customers to understand how many customers we have and their total business.

- The metrics you will create are: ***Number of Customers*** **and** ***Customer Lifetime Value***. You will then display these metrics on a **map and box-and-whisker chart**, respectively.

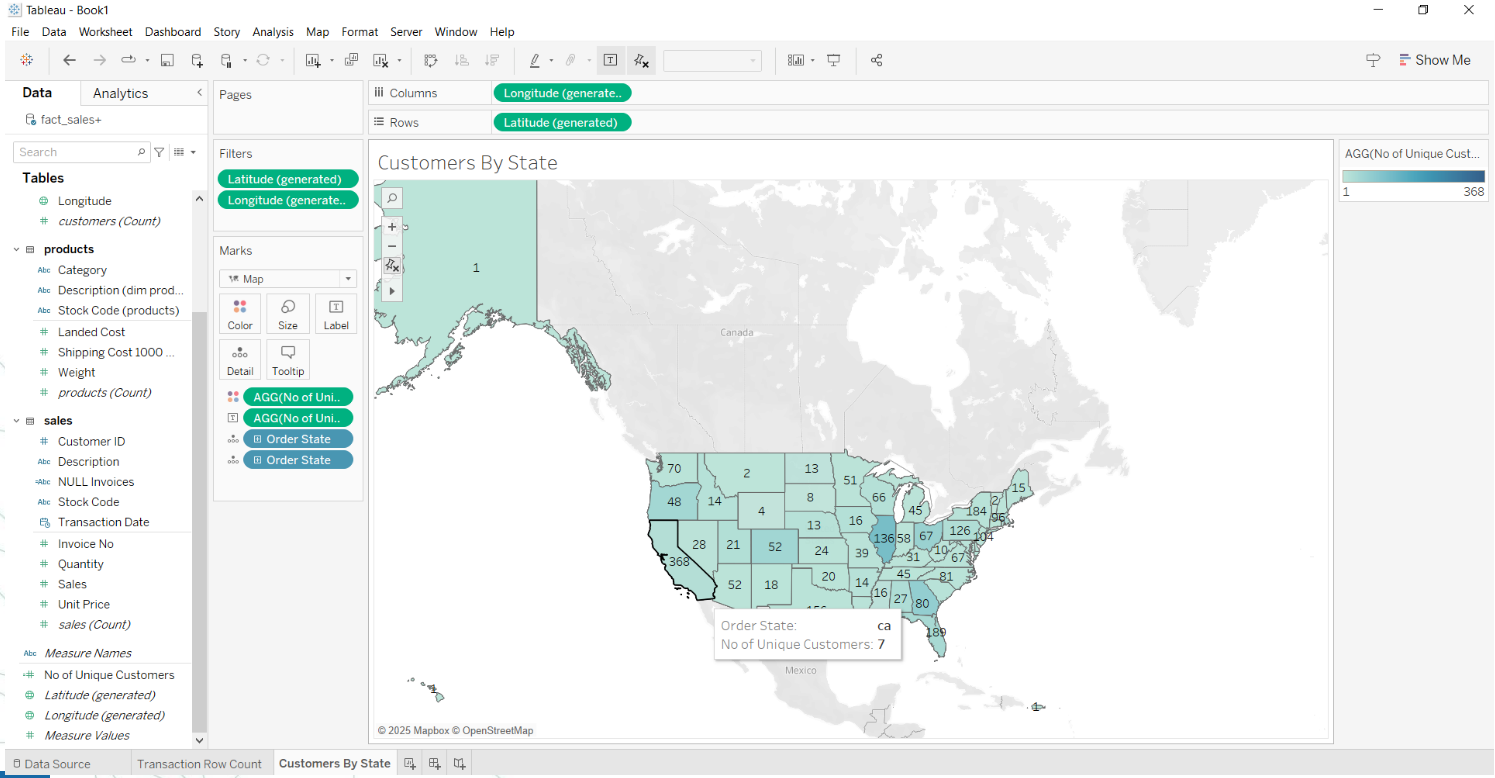
- **Create a calculated field name it "No of Unique Customers"** to count how many unique customers we have.

No of Unique Customers

```
COUNTD([Customer ID])
```

Step 8 – Building Metrics

- **Create a new sheet, "Customers by State", add a map, and display Number of Customers by state. Display the customer count as a text label on the state map. If you see Unknown locations (e.g., AE), filter those out. These represent overseas US military locations.**



Step 9 – Building Metrics

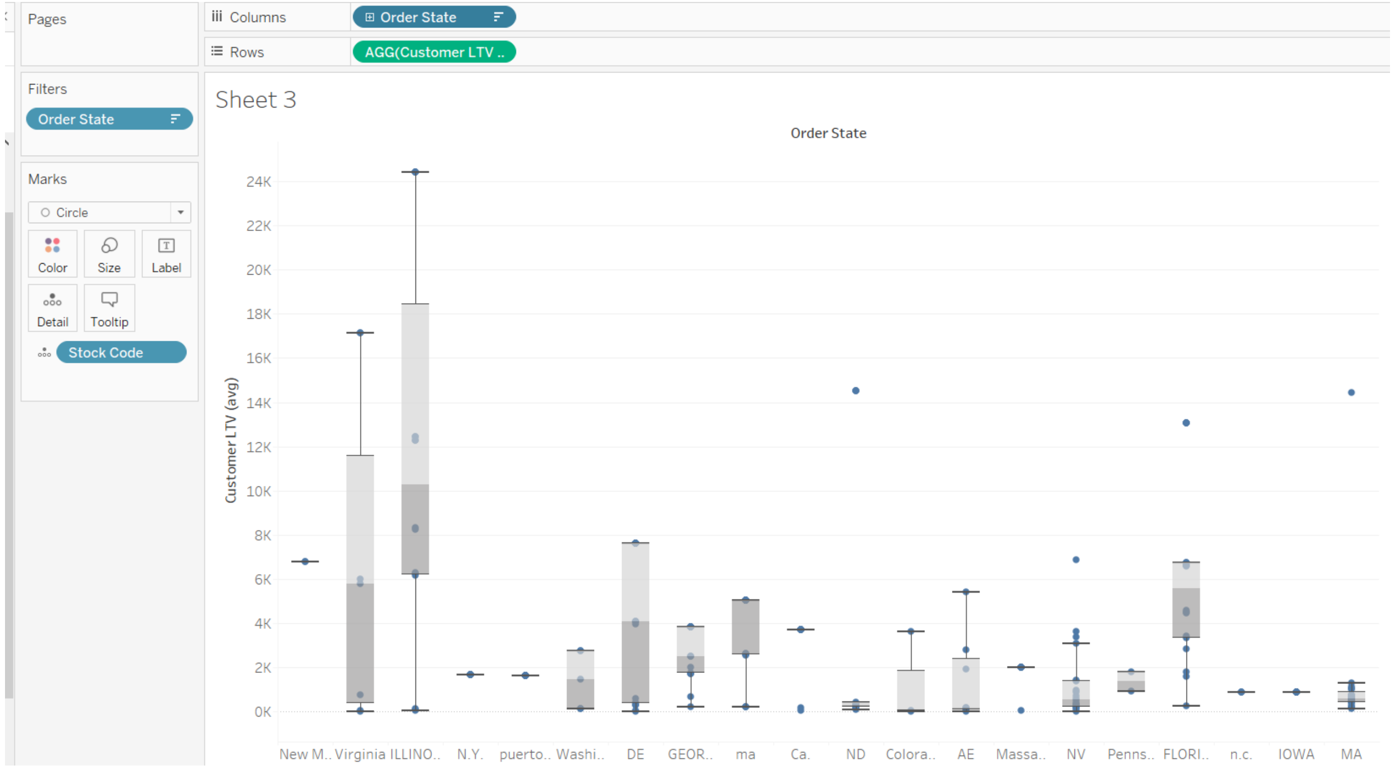
- Lifetime Value (LTV) means the total value of all sales by a customer.
- Create a field for "Customer LTV (avg)" and define it as the sum of all sales per customer. Use level of detail expression to calculate the value for each customer.

Customer LTV (avg)

```
AVG({FIXED [Customer ID] : SUM([Sales]) })
```

Step 10 – Building Metrics

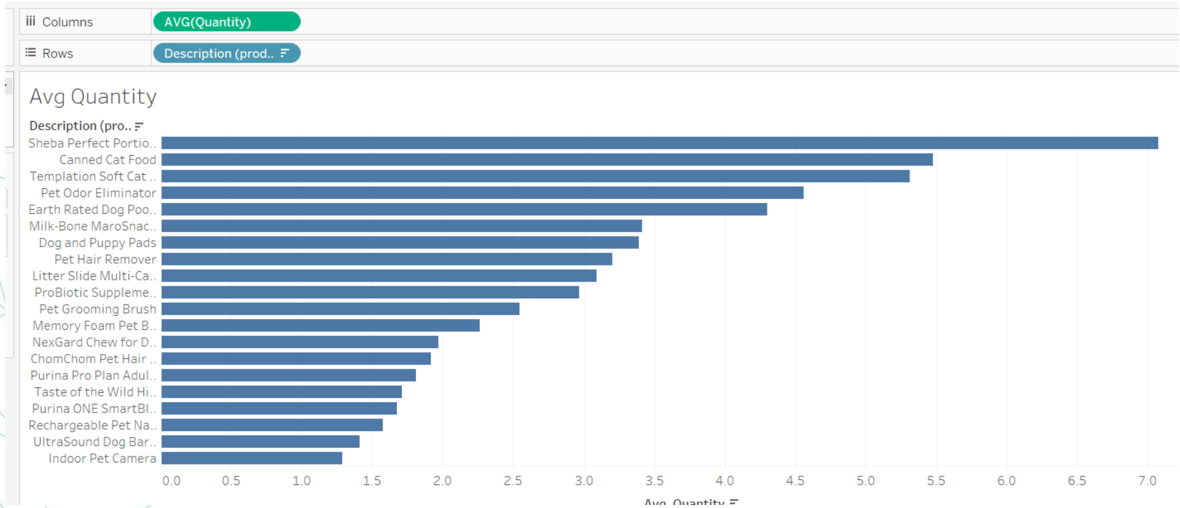
- Create a new sheet "LTV by Order State" and add a text table of Order State, Stock Code (from sales) , and Customer LTV.
- Convert the text table into a box and whisker plot of lifetime value (LTV) by Order State.
- Keep the top 20 states by LTV and sort descending on LTV.
- **Hoe much is the median for Virginia?**
- **If it is 5,772 you are on the right track , please proceed.**



Step 11 – Building Metrics

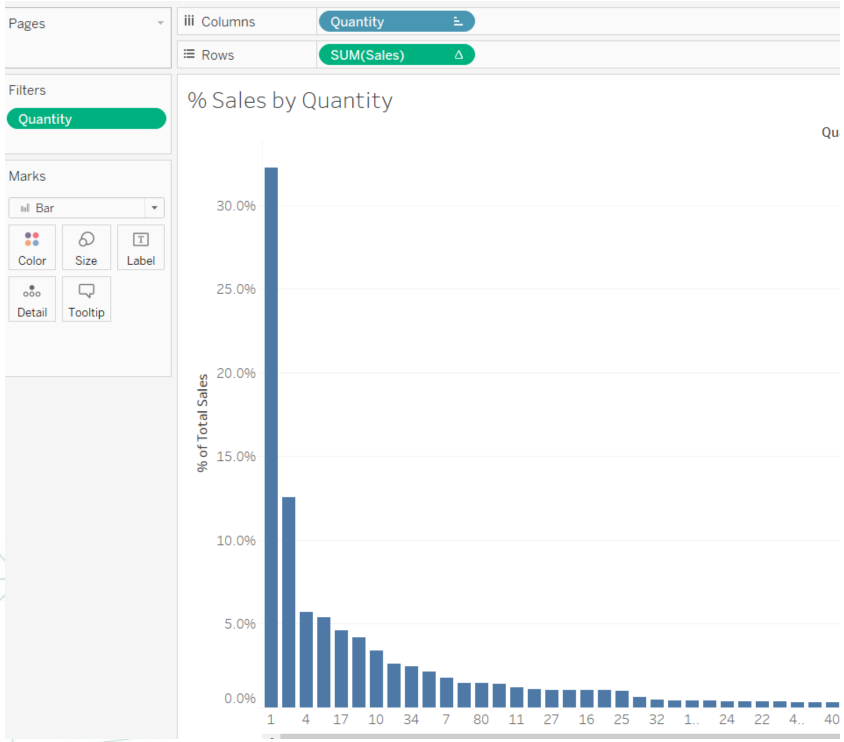
- Let us view the total sales by average quantity.
- Many customers buy a single quantity of a product but combine that with other product types.
- The average quantity differs when calculated at the invoice level vs. across the entire business.
- This step will solidify your understanding of the relationship between the shipped quantity and the shipping cost.

- **Create a new sheet "Average Quantity" and add a bar chart of average quantity by Description (products).**
- **Sort descending on average quantity.**



Step 12 – Building Metrics

- You want to understand the distribution of sales by quantity of a product across all sales.
- Create a new sheet, "% Sales by Quantity", and add a bar chart of the percentage of total sales by quantity.
- Round the sales percent of the total calculation to two decimals and filter out quantities less than zero. These represent returned orders.
- **Note you need to convert quantity to dimension on the shelf and the axis to discrete.**

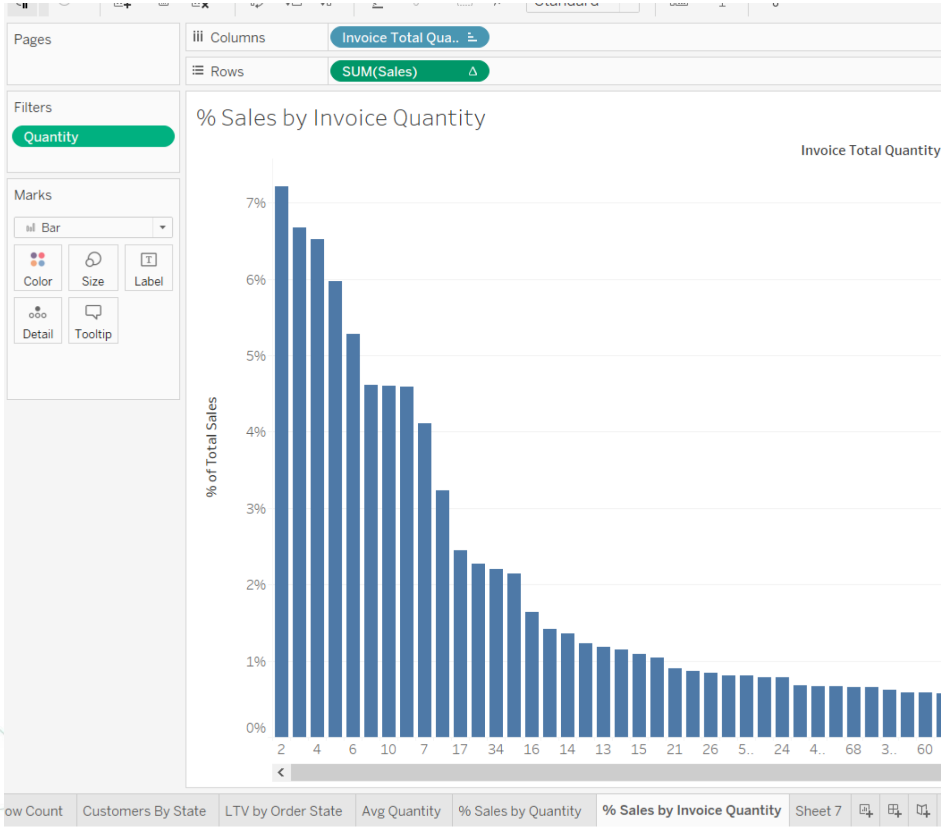


Step 13– Building Metrics

- Duplicate the sheet from the prior step and name it "% Sales by Invoice Quantity".
- create a new calculated field, “Invoice Total Quantity”.
- Define it as the sum of Quantity for each Invoice No.
- Make a note of the percentage of total sales where quantity equals one.
- **If your answer is 5.98% You are on the right track please proceed to the next step.**

Invoice Total Quantity

```
{FIXED [Invoice No] : SUM([Quantity])}
```



Step 14– Building Metrics

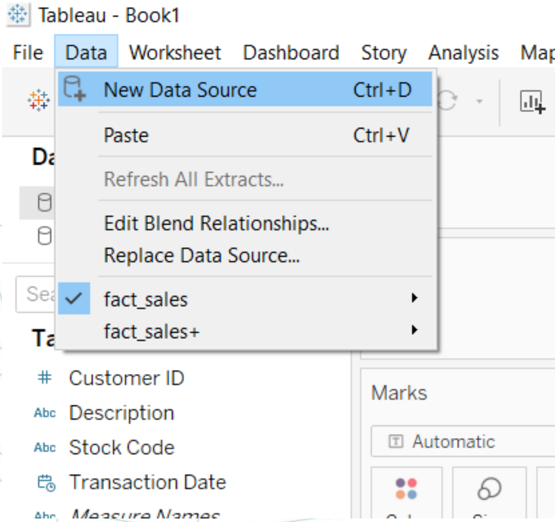
- The Shipping Cost 1000 mile field is the average cost to ship a single quantity of a product by itself (without combining it with other products).
- Create a new worksheet "per Unit Shipping Costs" and add a bar chart of **the average Shipping Cost 1000 mile by Description (products).**
- Sort descending on average Shipping Cost 1000 mile.

Step 14– Building Metrics

- The Shipping Cost 1000 mile field is the average cost to ship a single quantity of a product by itself (without combining it with other products).
- Create a new worksheet "per Unit Shipping Costs" and add a bar chart of **the average Shipping Cost 1000 mile by Description (products).**
- Sort descending on average Shipping Cost 1000 mile.

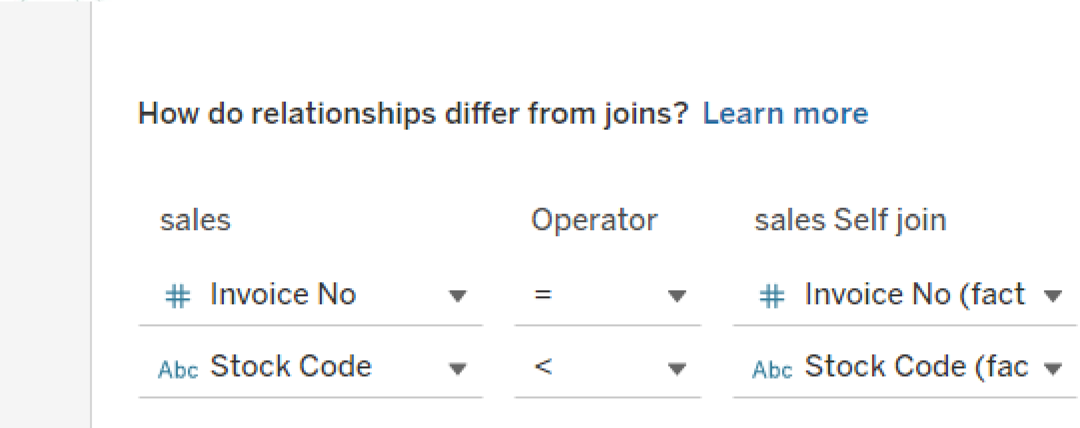
Step 15 – Market Basket Analysis

- The management wants to understand which products are frequently bought together.
- To find this out, you will build a correlation between the purchased quantity of products.
- You will connect with a transactions data CSV file and then connect the transactions table with itself (self-join).
- A self-join (on Invoice No) helps us understand products bought together as part of the same checkout.
- Create a **new** data source, **"Market Basket"**, based on the CSV file **fact_sales.csv**, which you'll use for the rest of this exercise. Feel free to rename the table with a user-friendly business name. For example, **sales**.



Step 16 – Market Basket Analysis

- Add the sales table (for a second time) and rename it "sales-self-join".
- Join the two tables. The first join criteria is on Invoice No, and the second is on Stock Code.
- Change the join criteria on Stock Code to the less than operator to reduce redundant stock code combinations. For example, in { (A,B), (B,A) }, we can eliminate one of the pairs.



Step 17 – Market Basket Analysis

- Define a calculated field Null Invoices to identify if there are records with no invoices in the sales or sales-self-join table.
- Define Stock Code Filter when Stock Code from either table has the same value.
- This represents exactly the same record in either of the transaction table.

Nulls

Market Basket

```
ISNULL([Invoice No]) OR ISNULL([Invoice No (fact sales.csv1)])
```

- Define **Stock Code Filter** when **Stock Code** from either table has the same value. This represents exactly the same record in either of the transaction table.

Stock Code Filter

Market Basket

```
[Stock Code] <> [Stock Code (fact sales.csv1)]
```

Step 18 – Market Basket Analysis

- Create a new sheet, "Rowcount Sales", and display the row count from the sales table.
- Make a note of the number of records.
- Apply a data source filter on Null Invoices and keep the "false" value.
- Similarly, apply a data source filter on State Code Filter to keep the "true" value.
- Make a note of the row count value from the sales table again.
- **What is the count of sales after applying the filters? If 12946 please proceed**

Step 19 – Market Basket Analysis

- Due to the self-join on the sales table, we have two copies of all attributes.
- Let us standardize attribute names and values.
- The market basket data source has two tables. The second instance of the table is renamed as **sales-self-join**. But the fields from the second table are not renamed.

•Create a new sheet, **"Product Description Variations"**.

•Create a calculated field **Product Description (self-join)** Create a similar field, **Product Description (sales).**

ıct Description (self-join) Market Basket

```
if [Description (self join)] = "Indoor Pet Camera (Wi-Fi)"
then "Indoor Pet Camera"
ELSE [Description (self join)] END
```

ıduct Description (Sales) Market Basket

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"
then "Indoor Pet Camera"
ELSE [Description] END
```

Step 20– Market Basket Analysis

- Due to the self-join on the sales table, we have two copies of all attributes.
- Let us standardize attribute names and values.
- The market basket data source has two tables. The second instance of the table is renamed as **sales-self-join**. But the fields from the second table are not renamed.

•Create a new sheet, **"Product Description Variations"**.

•Create a calculated field **Product Description (self-join)** Create a similar field, **Product Description (sales).**

ıct Description (self-join) Market Basket

```
if [Description (self join)] = "Indoor Pet Camera (Wi-Fi)"
then "Indoor Pet Camera"
ELSE [Description (self join)] END
```

ıduct Description (Sales) Market Basket

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"
then "Indoor Pet Camera"
ELSE [Description] END
```

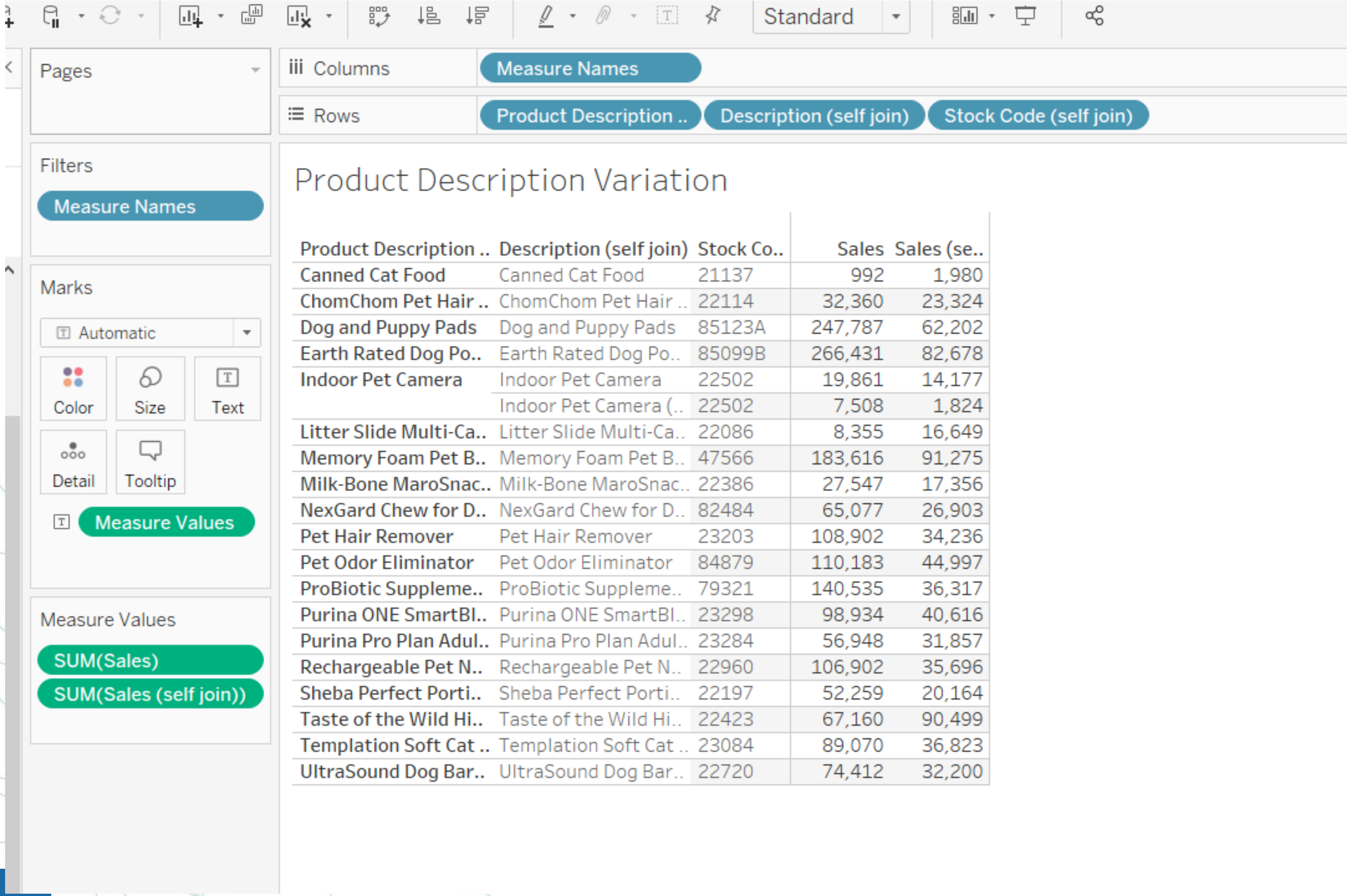
Step 21– Market Basket Analysis

- Switch to the old data source
- Create a new calculated field “product description and do the same as the last slide

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"
then "Indoor Pet Camera"
ELSE [Description] END
```

Step 22– Market Basket Analysis

- Generate this table and try your best to interpret the results?



Product Description ..	Description (self join)	Stock Co..	Sales	Sales (se..
Canned Cat Food	Canned Cat Food	21137	992	1,980
ChomChom Pet Hair ..	ChomChom Pet Hair ..	22114	32,360	23,324
Dog and Puppy Pads	Dog and Puppy Pads	85123A	247,787	62,202
Earth Rated Dog Po..	Earth Rated Dog Po..	85099B	266,431	82,678
Indoor Pet Camera	Indoor Pet Camera	22502	19,861	14,177
	Indoor Pet Camera (..	22502	7,508	1,824
Litter Slide Multi-Ca..	Litter Slide Multi-Ca..	22086	8,355	16,649
Memory Foam Pet B..	Memory Foam Pet B..	47566	183,616	91,275
Milk-Bone MaroSnac..	Milk-Bone MaroSnac..	22386	27,547	17,356
NexGard Chew for D..	NexGard Chew for D..	82484	65,077	26,903
Pet Hair Remover	Pet Hair Remover	23203	108,902	34,236
Pet Odor Eliminator	Pet Odor Eliminator	84879	110,183	44,997
ProBiotic Suppleme..	ProBiotic Suppleme..	79321	140,535	36,317
Purina ONE SmartBl..	Purina ONE SmartBl..	23298	98,934	40,616
Purina Pro Plan Adul..	Purina Pro Plan Adul..	23284	56,948	31,857
Rechargeable Pet N..	Rechargeable Pet N..	22960	106,902	35,696
Sheba Perfect Porti..	Sheba Perfect Porti..	22197	52,259	20,164
Taste of the Wild Hi..	Taste of the Wild Hi..	22423	67,160	90,499
Templation Soft Cat ..	Templation Soft Cat ..	23084	89,070	36,823
UltraSound Dog Bar..	UltraSound Dog Bar..	22720	74,412	32,200

Step 23– Market Basket Analysis

- Find correlations between **Quantity** and **Quantity (self-join)**.
- Create a calculated field “**Correlation**”.
- You need to use the **INCLUDE()** and **CORR()** functions.
- Including **Invoice No** in detail ensures that the correlation is calculated at the individual checkout level.

```
CORR(
 { INCLUDE [Invoice No]: SUM([Quantity]) },
 { INCLUDE [Invoice No]: SUM([Quantity (self-join)]) }
)
```

- Create a new sheet, "Market Basket Visualization", and display the correlation matrix by Stock Code and Stock Code (self-join) as a Highlight Table.
- Round Correlation to one decimal point and add Correlation and Product Description (self-join) to the table.

Step 24– Market Basket Analysis

- Enhance your visualization by hiding Stock Code (self-join), change colors (optional), and add a legend and a tooltip with the product description field from both sales and sales-self-join.
- Create a "Market Basket Analysis" dashboard and add the Market Basket Visualization sheet to this dashboard.
- **What is the correlation between pet hair remover (23203) and indoor pet camera (22502)? If -0.1 process**

Market Basket Visualization

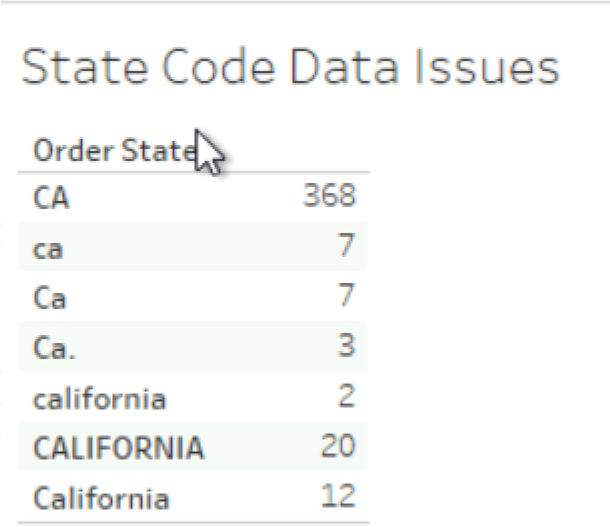
	Stock Code																
Product Description (self-join)	20725	21137	22086	22114	22197	22386	22423	22502	22720	22960	23084	23203	23284	23298	47566	79321	824
Canned Cat Food	0.4																
Litter Slide Multi-Cat Scented C..	0.4	0.3															
ChomChom Pet Hair Remover - ..	0.6	0.2	0.3														
Sheba Perfect Portions Pat We..	0.4	0.6	0.9	0.3													
Milk-Bone MaroSnacks Dog Tre..	0.7	0.9	0.5	0.1	0.6												
Taste of the Wild High Prairie G..	0.5	0.3	0.5	0.6	0.3	0.5											
Indoor Pet Camera	0.6		0.8	0.0	0.3	0.1	0.4										
UltraSound Dog Barking Deterr..	0.9	0.0	0.1	0.7	0.5	0.6	0.9	0.5									
Rechargeable Pet Nail Grinder	0.9	1.0	0.4	0.6	0.5	0.3	0.6	0.2	0.8								
Templation Soft Cat Treats	0.6	0.0	0.8	0.3	0.8	0.9	0.4		0.6	0.5							
Pet Hair Remover	0.8	0.0	0.9	0.9	0.8	0.7	0.8	-0.1	0.5	0.5	0.8						
Purina Pro Plan Adult Sensitive..	0.6	0.2	0.2	0.3	0.3	0.3	0.9	0.3	0.3	0.9	0.0	0.9					
Purina ONE SmartBlend Natura..	0.6	0.1	0.3	0.2	0.2	0.5	0.5	0.2	0.6	0.2	0.0	0.6	0.4				
Memory Foam Pet Beds for Sm..	0.6	0.2	0.2	0.2	0.2	0.7	0.4	0.5	0.6	0.6	0.0	0.7	0.3	0.7			
ProBiotic Supplements for Dogs	0.6	0.2	0.4	0.3	0.4	0.5	0.4	0.7	0.2	0.3	0.3	0.4	0.4	0.1	0.4		
NexGard Chew for Dogs	0.7		0.7	0.1	0.7	0.3	0.4	0.9	0.8	0.3	0.0	0.8	0.6	0.6	0.3	0.6	
Pet Odor Eliminator	0.6	0.8	0.4	0.5	1.0	0.8	0.7	0.4	0.7	0.6	0.7	0.9	0.2	0.3	0.3	0.5	
Earth Rated Dog Poop Bags	0.8	0.7	0.4	0.2	0.6	1.0	0.5	0.3	0.6	0.7	0.3	0.9	0.5	0.7	0.6	0.7	
Dog and Puppy Pads	0.7	0.1	0.5	0.2	0.6	0.7	0.3	0.4	0.4	0.5	0.2	0.7	0.4	0.5	0.5	0.6	

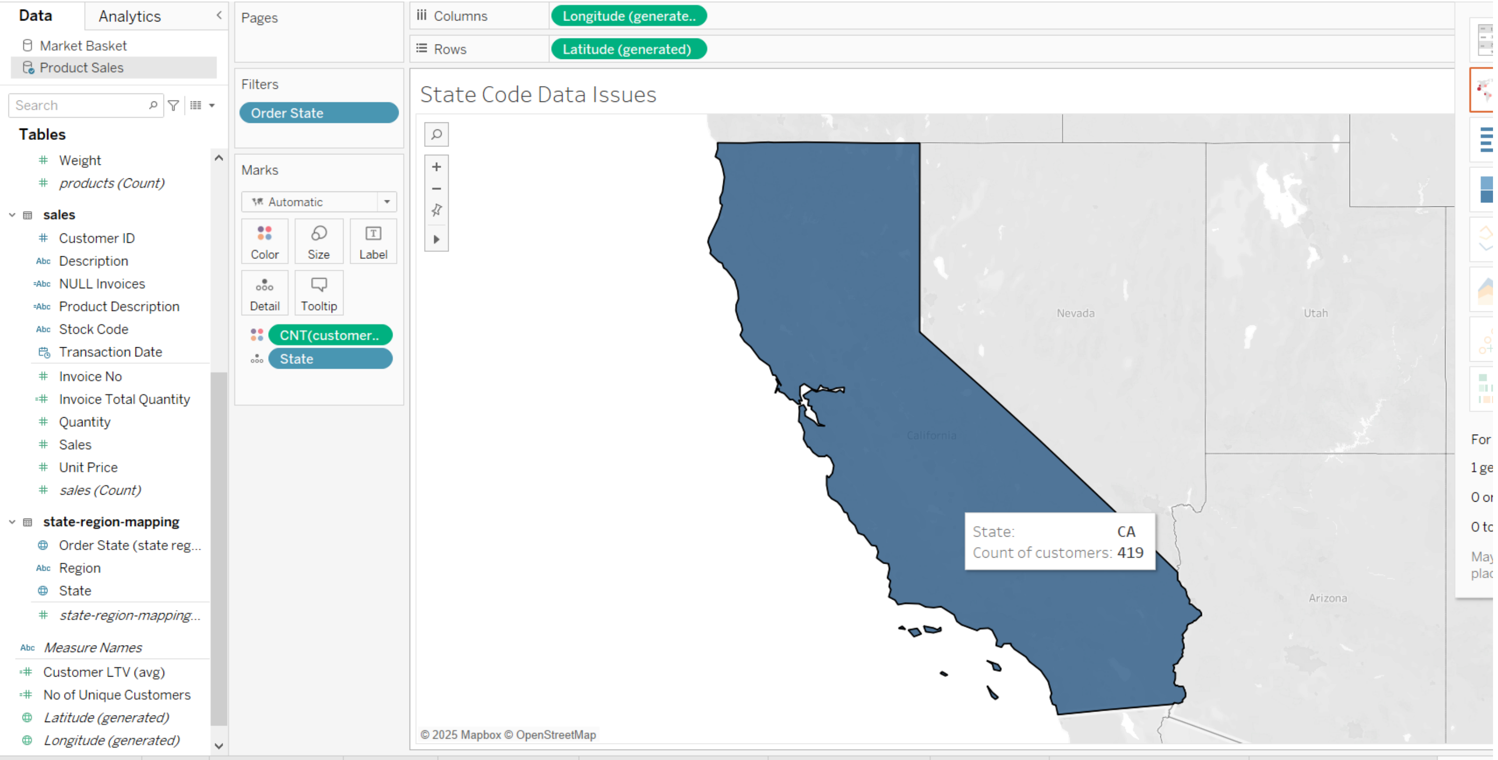
Step 25– Shipping Cost Analysis

Step 25– Shipping Cost Analysis

- Tableau recognizes geographical elements such as country and state automatically.
- There are limitations if your geographic attributes have multiple variations of values.
- We will look at a specific example of this limitation and devise a solution using a custom state mapping in the data source.
- You will be working with the Product Sales data source.
- **Add a new sheet, "State Code Data Issues".**
- Create a text table showing the number of customers by Order State.
- Filter Order State and keep all variations of the state name "California" or state code "CA".
- Make a note of the number of customers in "CA" and the total number of customers for all different variations of state code "CA" or state name "California".
- Change the text table to a map.
- Tableau only recognized the state code **CA**, leaving the other six records unmapped.
- You can fix this by adding a custom state mapping.
- Add **state-region-mapping.csv** from the *Datasources* folder to the **Product Sales** data source and rename the CSV table to **"state-region-mapping"** and join on the state in the **Customers** table.
- Replacing **Order State** with **State** from the custom mapping will cause unmapped variations of California to map correctly to the state. "Unknown" and "AE" locations refer to overseas US military so ignore it.

Step 26– Shipping Cost Analysis

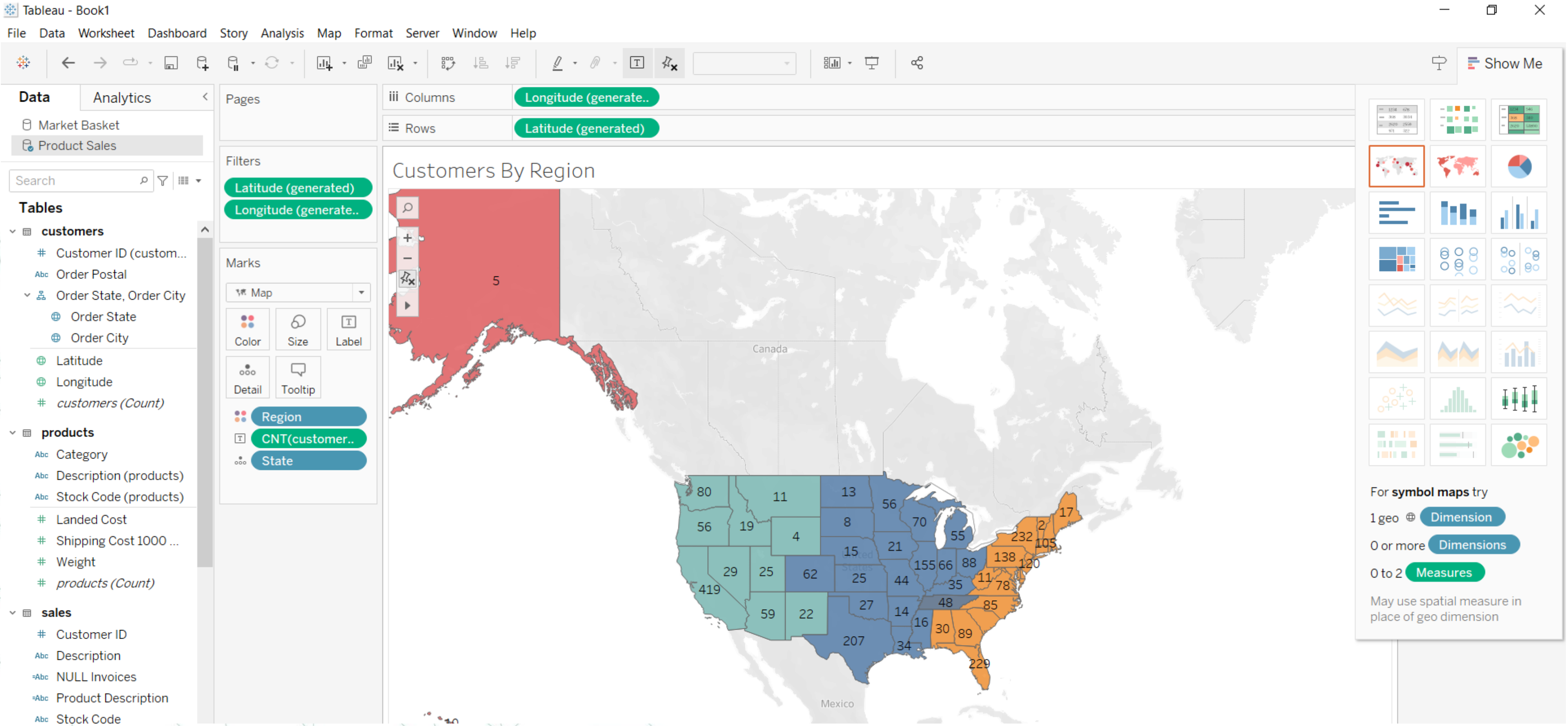




Step 27– Shipping Cost Analysis

- Duplicate the sheet **Customers by State** created previously and rename it **"Customers by Region"**.
- Move this new sheet to the farthest right in the workbook.
- Replace **Order State** with **State** and add color by **Region** and **Number of Customers** as a data label.
- **How many additional records are associated with the state of California after you added the custom region mapping? If your answer is 51 please proceed.**

Step 27– Shipping Cost Analysis



Step 28– Shipping Cost Analysis

- Management wants to understand where its customers are and the total sales by state.
- To build this map, you will use spatial functions and build a **distance** metric.
- The **makepoint()** function creates geometric points on earth while **distance()** calculates the distance between two locations on earth.
- The store has a warehouse in Los Angeles, CA. The latitude and longitude are **(34.0544, -118.2439)**. Using Tableau's spatial functions, create a spatial object or a geographic point, **"Warehouse-LAX"**.
- Make a calculated field "Warehouse-LAX" as the following.

Warehouse-LAX   Product Sales

```
MAKEPOINT(34.0544, -118.2439)
```

- Every customer address includes latitude and longitude. This is the destination for each order.
- Create a spatial field **"Destination"** using the customer's latitude and longitude.
- MAKEPOINT([Customer Latitude], [Customer Longitude])

Destination   Product Sales

```
MAKEPOINT([Latitude], [Longitude])
```

Step 29– Shipping Cost Analysis

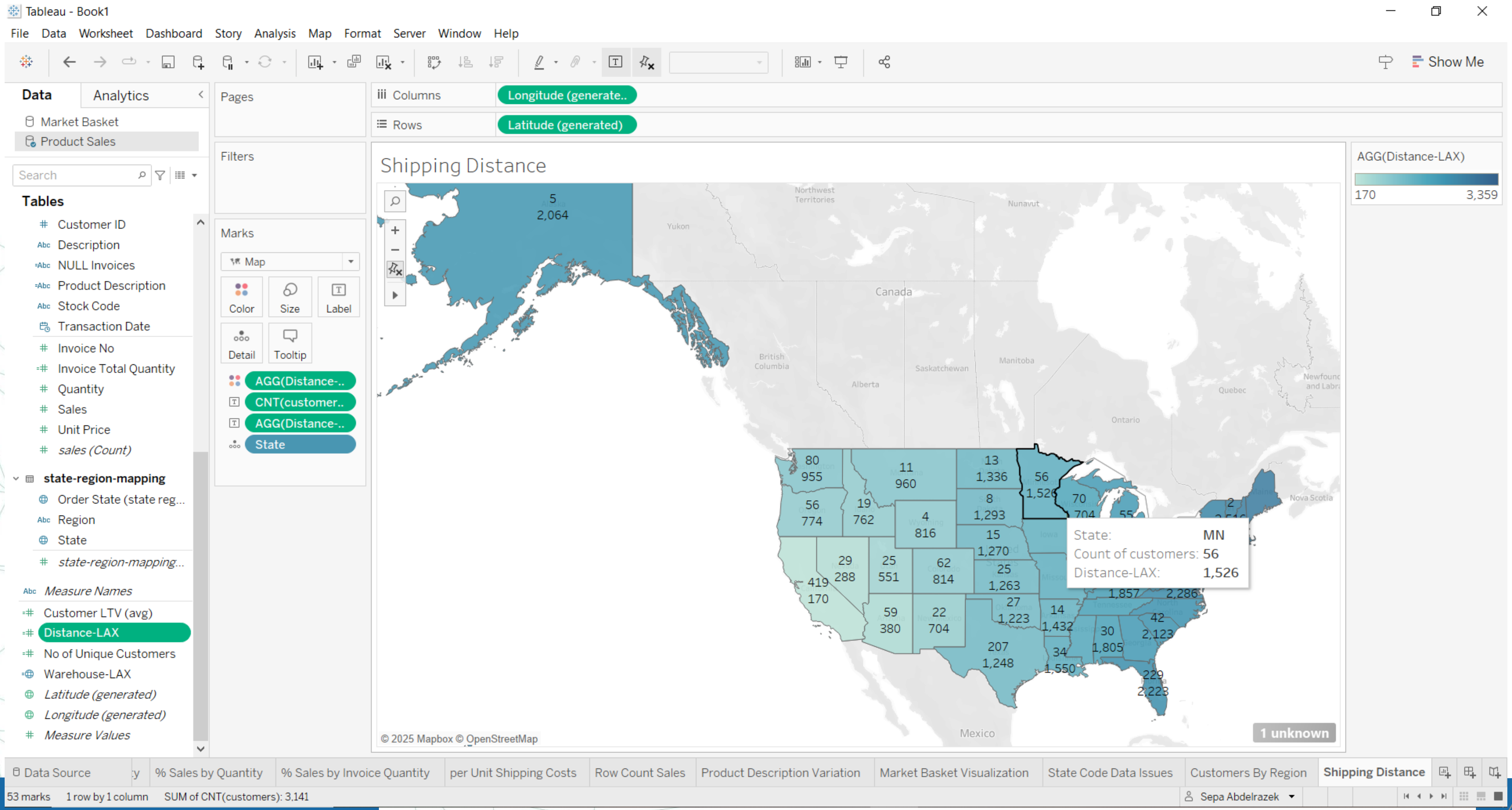
- You now have two geographic or spatial fields defined.
- Calculate the distance between these two points as **"Distance-LAX"**.
- Use average aggregation and miles as the unit.

Distance-LAX   Product Sales

```
avg(DISTANCE([Warehouse-LAX], [Destination], "mi"))
```

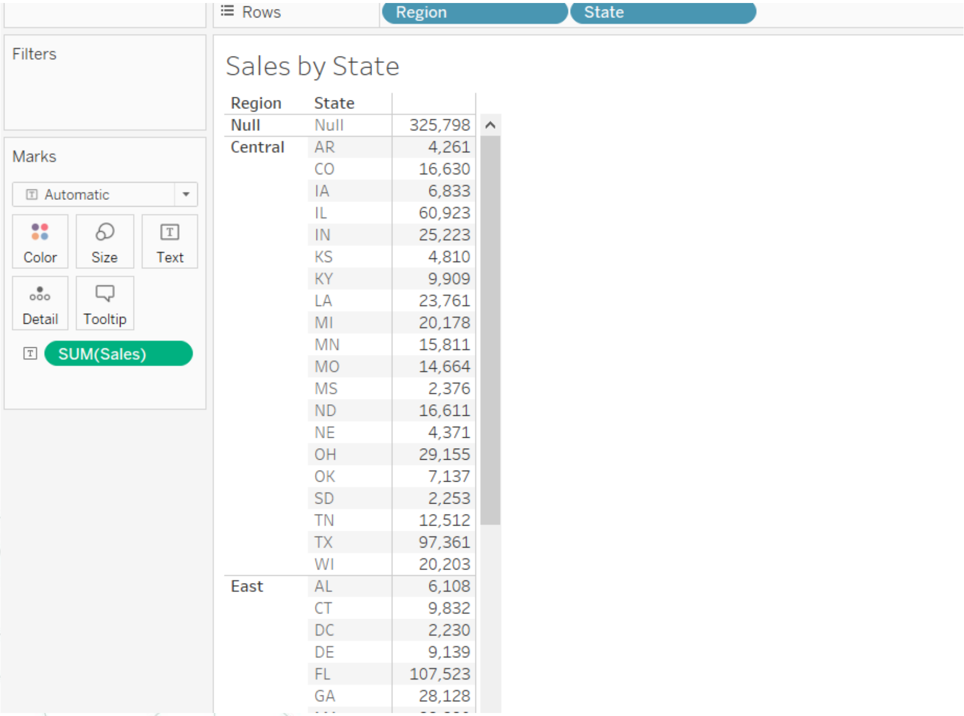
- Add a new sheet, **"Shipping Distance"**, and create a map of the US and color states by the average distance to the LAX warehouse.
- Add the number of customers and the distance to the tooltip.
- **What is the average shipping distance in miles between New York and the Los Angeles (LAX) warehouse?**
- If your answer is 2426 then proceed

Step 29– Shipping Cost Analysis



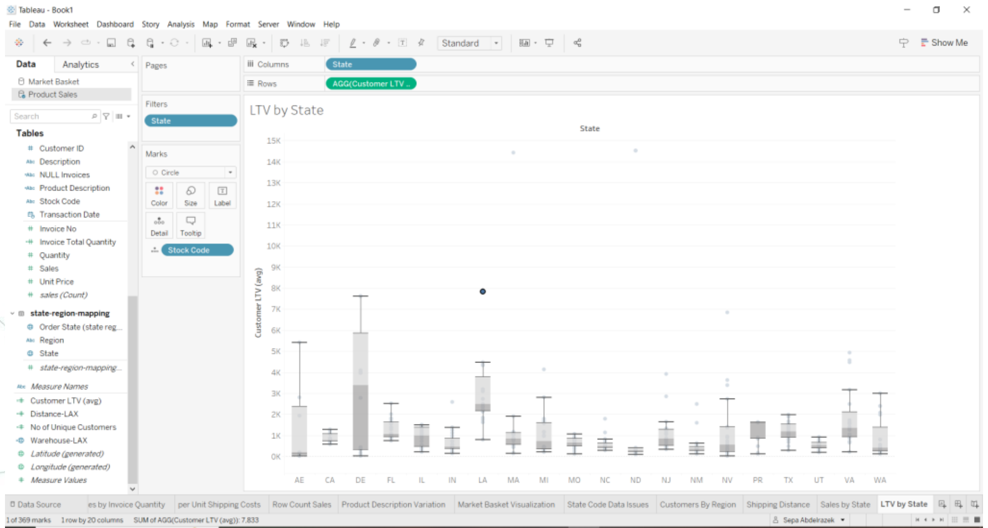
Step 30– Shipping Cost Analysis

- Create a new sheet, "Sales by State", and add a text table using State and Region dimensions from the state-region-mapping table.
- Add the Sales measure to the table.



Step 31– Shipping Cost Analysis

- We created a box-and-whisker plot of average **Customer LTV** by **Order State**.
- We will recreate this chart using the **State** field from **state-region-mapping** and see the impact of using the custom state mapping.
- Duplicate LTV by Order State (created in Before) as "LTV by State" and replace the old state fields with the new fields from the custom mapping.
- Move the duplicated and renamed sheet to the furthermost right. Remove the top 20 filter on Order State and replace it with the top 20 filter on State.
- **What is the median customer lifetime value in the state code VA? If 1366 proceed.**
- Note that, The custom mapping increased the record count in Virginia by **331**. This improved the accuracy of the median value from an earlier value of **5,772** and reduced the gap with the overall customer LTV of **1,366** across all states.



Step 32– Shipping Cost Analysis

- Due to a lack of automatic shipping cost capture at the transaction level, the shipping department relies on spreadsheets.
- The shipping department has told you that shipping more than one quantity of an item costs, on average, **70% of the cost of a single-unit shipment**. There are variances based on the product.
- You want to allow the users to play around with effective shipping rates as the quantity changes.

- Create a **"Shipping (Baseline)"** field. If the quantity is just a single item, then there is a regular shipping cost (**Shipping Cost 1000 mile**). If more than one item is shipped, the cost per unit shipped is **70% for an additional quantity above one**. For example, if five items are shipped, then the shipping is:

100% of the cost of the first unit
+
0.7 * ((5-1) * the cost of unit shipment)

Step 32– Shipping Cost Analysis

Shipping (Baseline)   Product Sales

```
if [Quantity] = 1 then [Shipping Cost 1000 mile]
ELSE
[Shipping Cost 1000 mile] + (([Quantity]-1 )* [Shipping Cost 1000 mile] * 0.7 )
END
```

Step 33– Shipping Cost Analysis

- **Create a parameter** What-if quantity of integer type and allow values from 1-20 in steps of one.
- Set the current value to **five**.
- **Create a calculated field**, Blended Shipping Cost Factor, which calculates the discounted shipping cost based on the following table.
- The parameter allows the user to try different values for effective shipping rates rather than using a fixed value of 0.7 used in step one.
- The formula below maps a set of parameter values to a fixed cost multiplier.

`What-if quantity`
<= (less than or equal), cost multiplier (output)
1, 1
2, 0.8
4, 0.6
7, 0.5
9, 0.4
else, 0.3

Step 34– Shipping Cost Analysis

- Duplicate the **Shipping (Baseline)** field and rename it to **"Shipping (What-if)"**.
- Use **Blended Shipping Cost Factor** in place of the hard-coded 0.7 value.
- Define the field **"Shipping (Difference Baseline)"** as the difference between **Shipping (Baseline)** and **Shipping (What-if)**.
- On a new sheet, **"Shipping Metrics"**, add the three shipping metrics to a text table.
- **What is the value of Shipping(What-if)? If 327,037 proceed**
- **Note that Total Shipping Cost decreased to 327,037 by when the avg shipped quantity is 5.**

Shipping Metrics

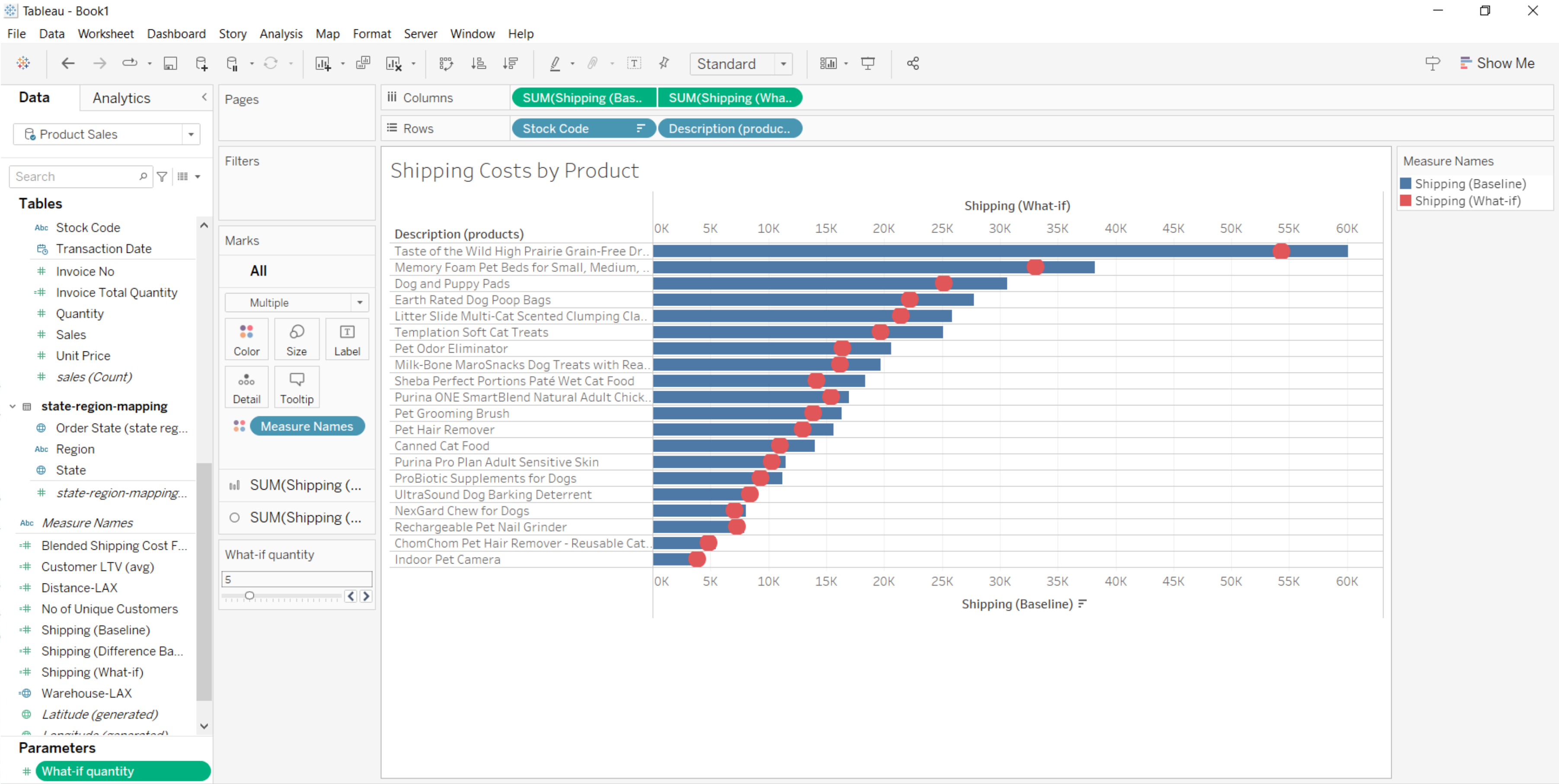
Shipping (Baseline)	386,525
Shipping (Difference Bas..	59,488
Shipping (What-if)	327,037

Step 35– Shipping Cost Analysis

- Using the fields created in the prior step, let us compare shipping costs (baseline) with a hypothetical what-if scenario.
- Change parameter values and see how it impacts shipping costs.
- On a new sheet, **"Shipping Costs by Product"**,
- create a **horizontal bar chart** displaying **Stock Code**, **Product Description**, and **Shipping (Baseline)** metrics.
- Add the Shipping (What-if) metric on a dual axis.
- Synchronize Axis and change the mark type of Shipping (What-if) to a Circle.
- Display the What-if Quantity parameter and observe how it impacts the measure values.
- **What will be the shipping cost savings (the difference between baseline and what-if shipping amount) for Dog and Puppy Pads when the What-if Quantity is set at five? If 5,344 proceed**
- That's correct. As per the **Blended Shipping Cost Factor**, when the shipped quantity is **five**, each quantity above one only costs **50% of the unit shipping cost**.
- The **Shipping (Baseline)** amount is **$30,492**.
- The **Shipping (What-if)** amount is **$25,148**.

Step 36– Shipping Cost Analysis

Adjust your visual to be the same.



Step 37– Shipping Cost Analysis

- The management wants to see a **dashboard** displaying shipping metrics and an ability to analyze what-ifs.

Here are the business requirements:

• **Prominently display key KPIs**. They are sales and cumulative shipping costs (baseline, what-if, and difference).

•**Change shipped quantity using a parameter** and an action filter on the product.

•**Breakdown of baseline and what-if shipping costs by product**.

- Display **key KPIs in a large font**. They are also called **Big Angry Numbers (BANs)**, and help answer important questions right away.

Your dashboard will need to display **four KPIs in large numbers**.

- Create **four new sheets** and visualize **sales** and the three main **shipping metrics** (baseline, what-if, and difference baseline).

• Name the sheets: **"BAN-Sales"**, **"BAN-Shipping Baseline"**, **“BAN-Shipping What-IF” and “BAN-Shipping Difference”**.

Step 38– Shipping Cost Analysis

Create a New Sheet - "Shipping What-if"

•Display the **running total** of the three main shipping metrics in an **area chart with transaction month**.

•Apply **color** to differentiate metrics.

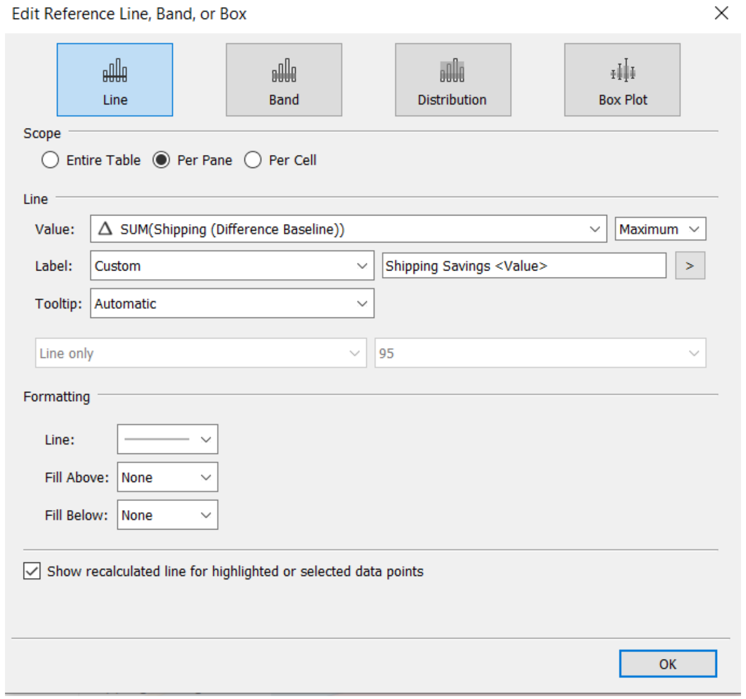
Add a Reference Line

•Insert a **reference line** on the **area chart**.

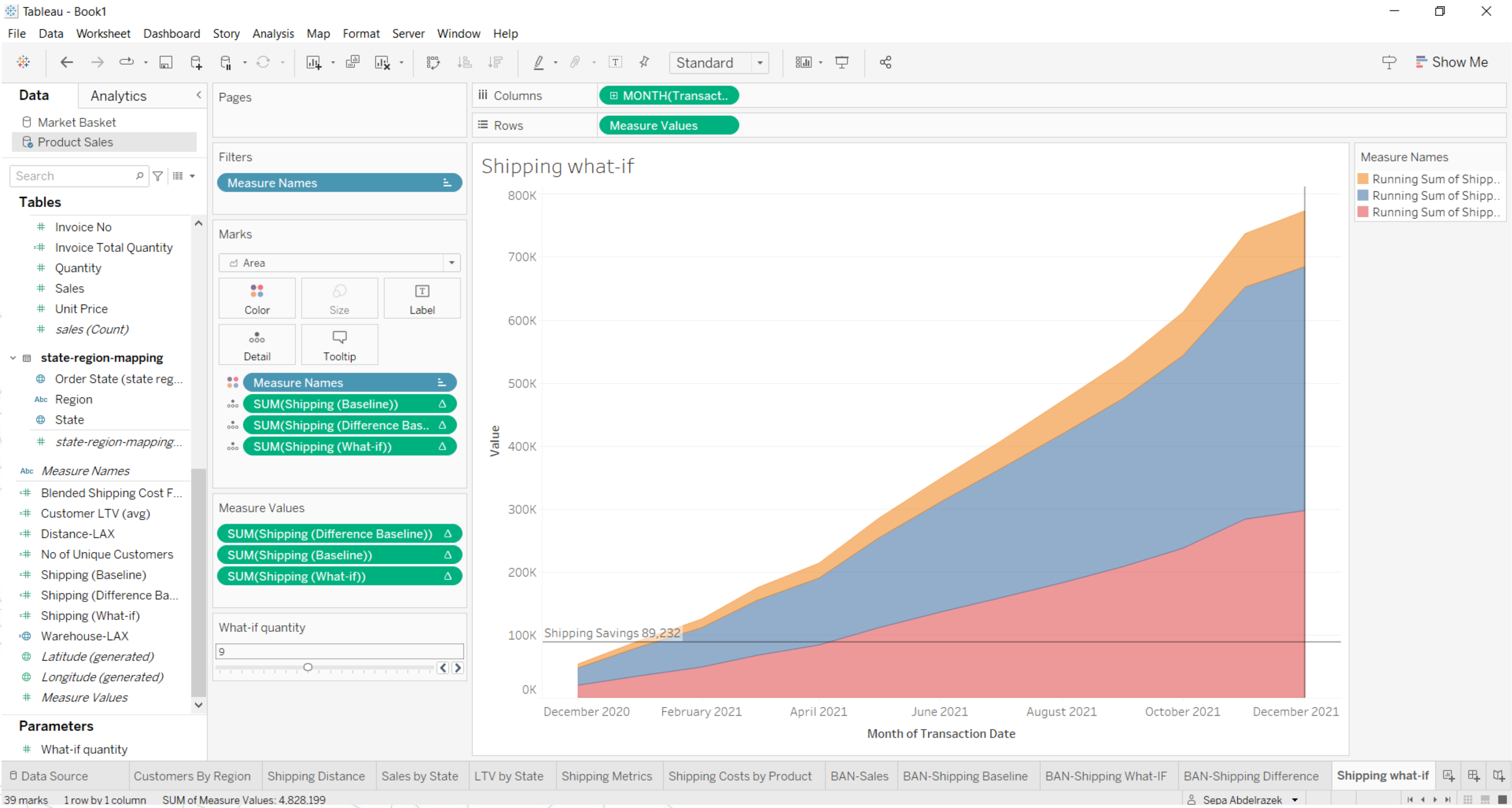
•Add a **custom label: "Shipping savings"**.

•Include the **running total of Shipping (Difference Baseline)** in the reference line.

•Please navigate to next slide to see how it should be



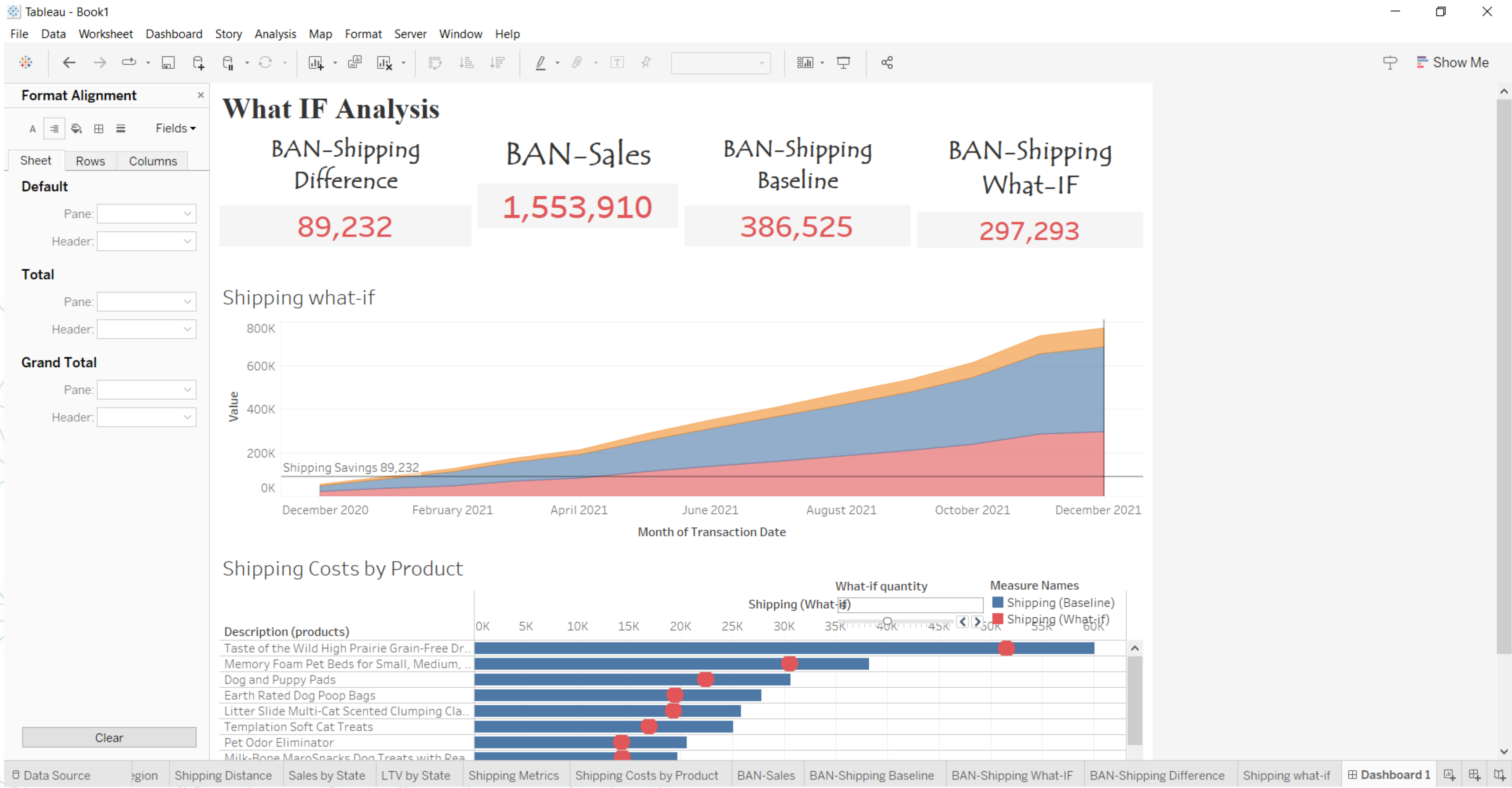
Step 39– Shipping Cost Analysis



Step 40– Shipping Cost Analysis

- **Create the What-If Analysis Dashboard**
- **Visualize the What-If Dashboard ("What-If Analysis")** using:
  - **Four big KPIs (BANs)**
  - **Area chart (Shipping What-if)**
  - **Shipping Costs by Product** sheet.
- Set the dashboard size to **at least 1200×1000** or as per preference.
- **Ensure the What-If parameter is displayed** and enable **Total Shipping Costs by Product** as an **action filter**.

Step 40– Shipping Cost Analysis



Step 41– Data Visualization & Storytelling

- Your data story will consist of three dashboards.
- You will reuse many of the charts built in the previously.
- In the **executive summary dashboard**, you will provide key KPIs such as sales and profit. This dashboard will provide a high-level pulse of the business.
- **The growth opportunities dashboard** will provide specific recommendations on how we can grow sales using cross-sell promotions.
- **The shipping costs dashboard** will suggest multiple strategies to reduce shipping costs.

Step 42– Data Visualization & Storytelling

- Revenue by location and top sellers
- Let us build core KPIs which executives frequently ask to see.
- For example, what are my top-selling products, and who are my top customers? To answer this, using the **Product Sales** data source, you will build a map of sales by state and a bar chart for top-selling products.
- **Create a new sheet, "Sales by State Map".**
- Display a map with states colored by the amount of sales and add the Sales amount as a label.
- Display the sales amount in thousands (integer), with the US dollar currency symbol.

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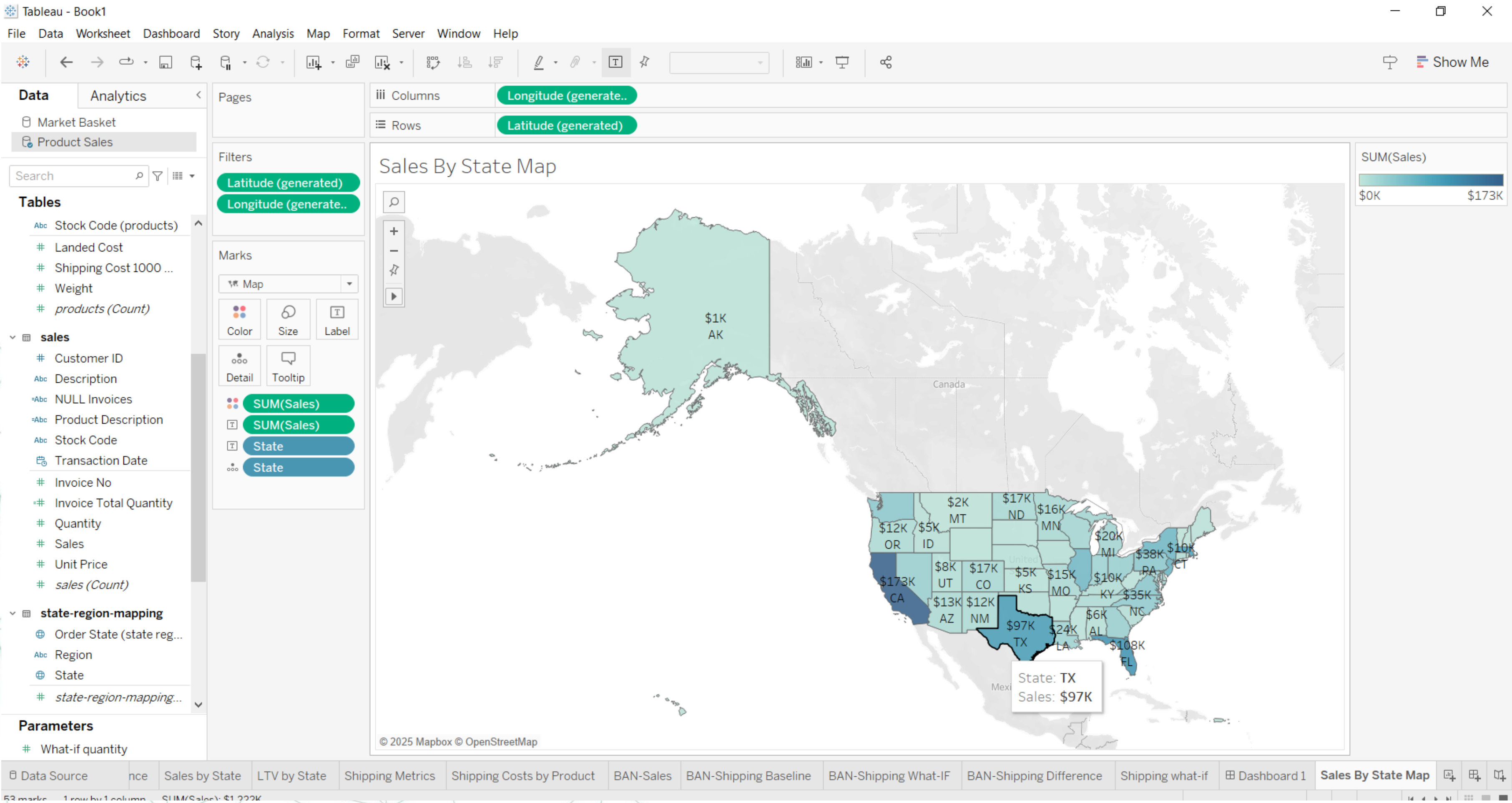
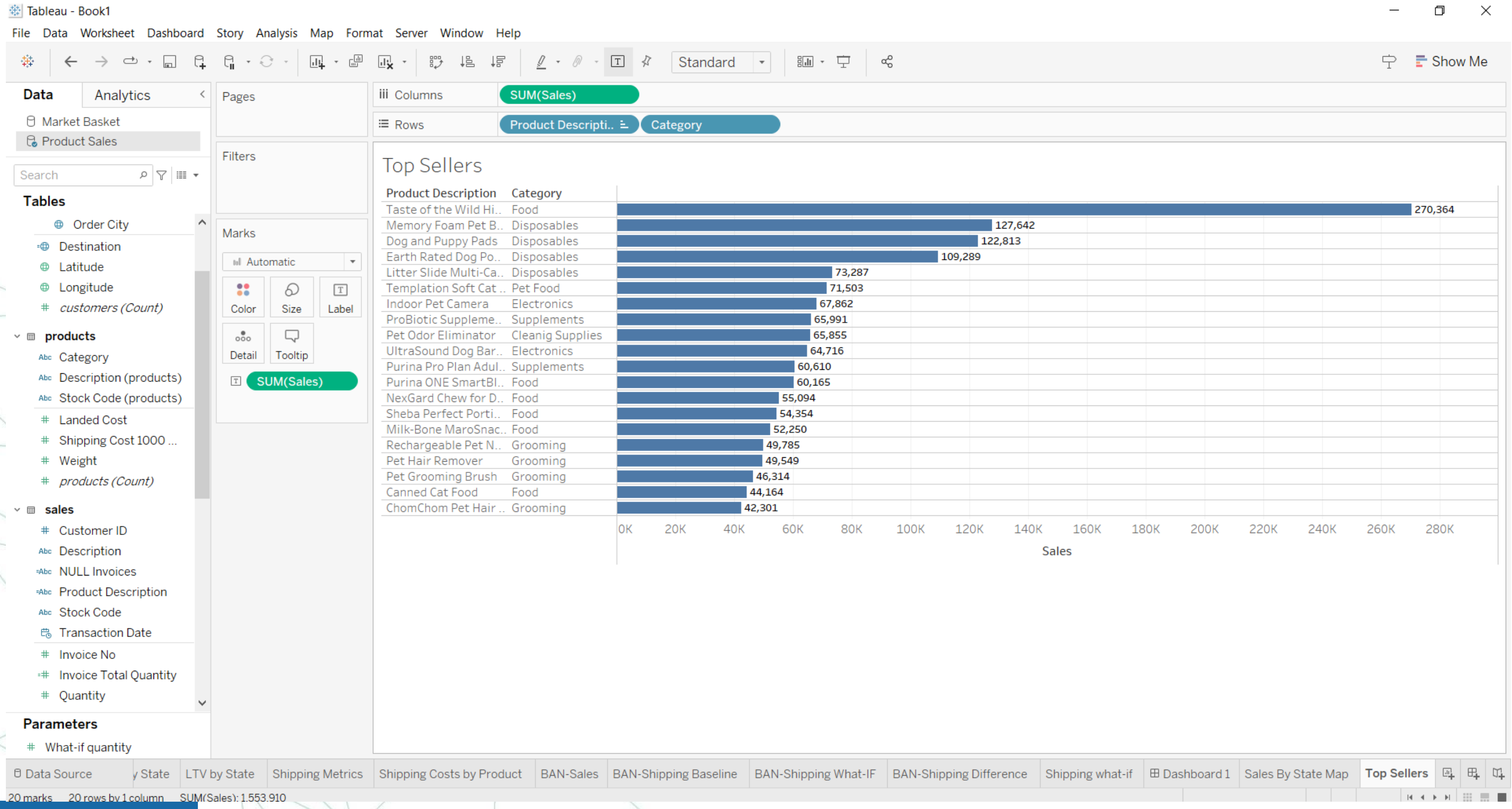
Step 42– Data Visualization & Storytelling



Step 43– Data Visualization & Storytelling

- **Create a new sheet, "Top sellers".**
- Create a horizontal bar chart of the amount of sales by Product Description and Category.
- Sort descending on the Sales amount and display the sales amount as a label.



Step 44– Data Visualization & Storytelling

- **KPIs in Big Angry Numbers (BANs)**
- You will be creating **BANs (Big Angry Numbers)** similar to those you created previously.
- You will build three different BANs: **Profit, Profit Percentage, and Shipping Baseline**.
- **Create two new calculated fields.**
- The First calculated field COGS (cost of goods sold) as the multiplication of Landed Cost and Quantity.
- **Create Profit (Baseline) as sales subtracted by COGS and Shipping (Baseline).**
- **Set the default** number format for both fields to a US dollar integer value.
- **Create a third field, Profit %, and define it as: the sum of Profit (Baseline) divided by the sum of Sales.**
- Set the number format as a percentage with two decimal points.
- **Create a new sheet, "BAN-Profit", and display Profit (Baseline) in a large font.**
- Format the sheet to remove all lines and borders.
- **Create two new sheets, "BAN-Profit Percentage" and "BAN-Shipping Baseline", by duplicating BAN-Profit and replacing: Profit (Baseline) with Profit %. Profit (Baseline) with Shipping (Baseline) respectively.**
- **What is the profit amount on the "BAN-Profit Percentage" sheet? If 27% you are on the right track**

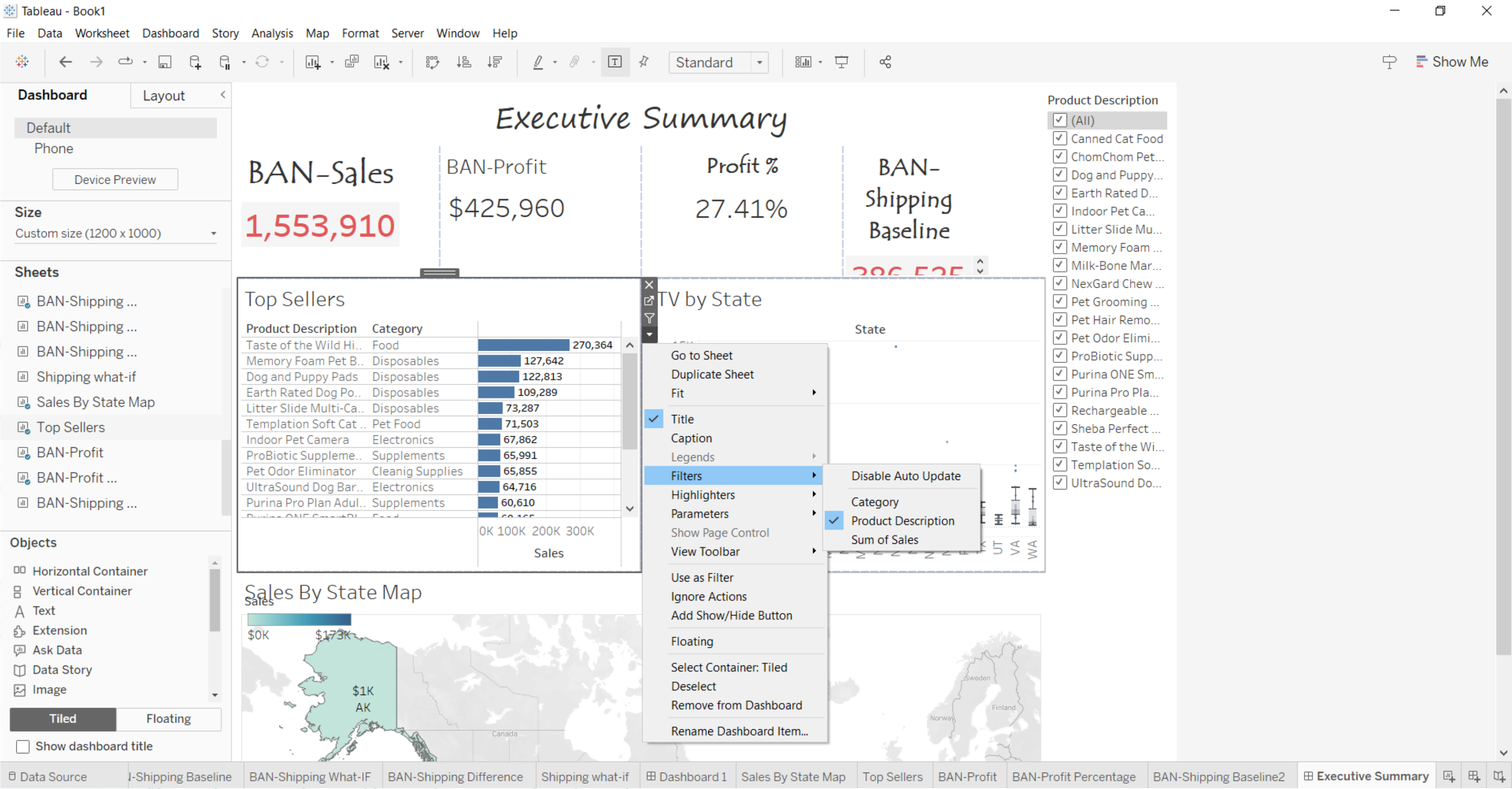
Step 45– Data Visualization & Storytelling

- **Build an Executive Summary.**

We now have all the pieces to create an executive summary. An executive summary provides a quick pulse of the business operations and displays metrics such as sales, profit, and expenses. In addition, it provides an ability to drill down or filter on dimensions like product and customer location.

- **Create a new dashboard**, **Executive Summary**, with a custom size of at least 1200px width and 1000px height.
- Add the **BAN-Sales, BAN-Profit, BAN-Profit Percentage, and BAN-Shipping Baseline**
- **Replace titles, align text, and ensure all four sheets are evenly aligned.**
- **Add Top sellers, Sales by State Map, and LTV by State to the dashboard.**
- Adjust titles, alignment, and formatting.
- Apply a filter on Product Description to all sheets on the dashboard and use the map as an action filter.

Step 45– Data Visualization & Storytelling



Step 46– Data Visualization & Storytelling

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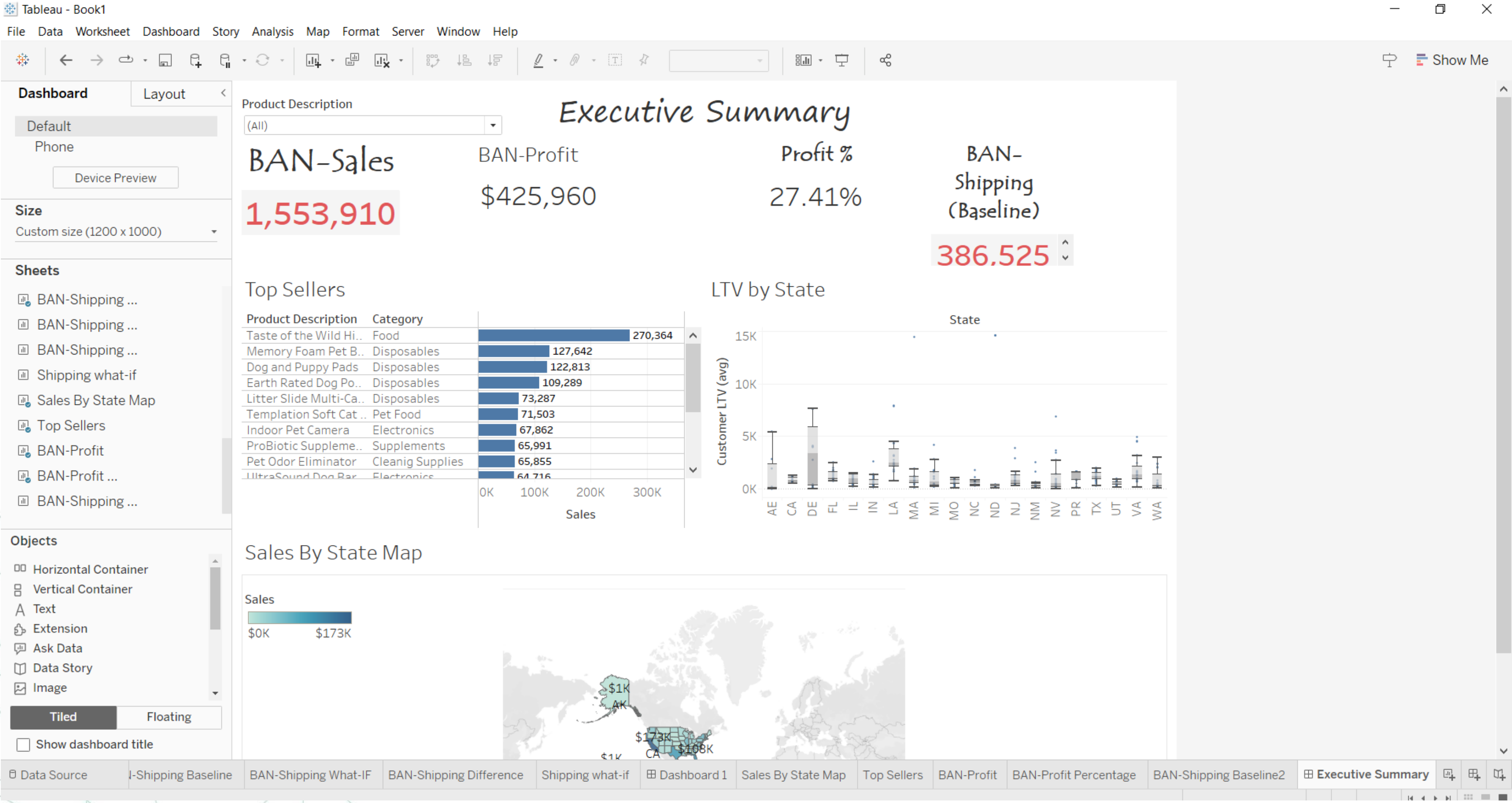
Align Your Dashboard

What is the profit % margin for Taste of the Wild High Prairie dog food in the state of California (code: CA)?

IF 26.26% proceed

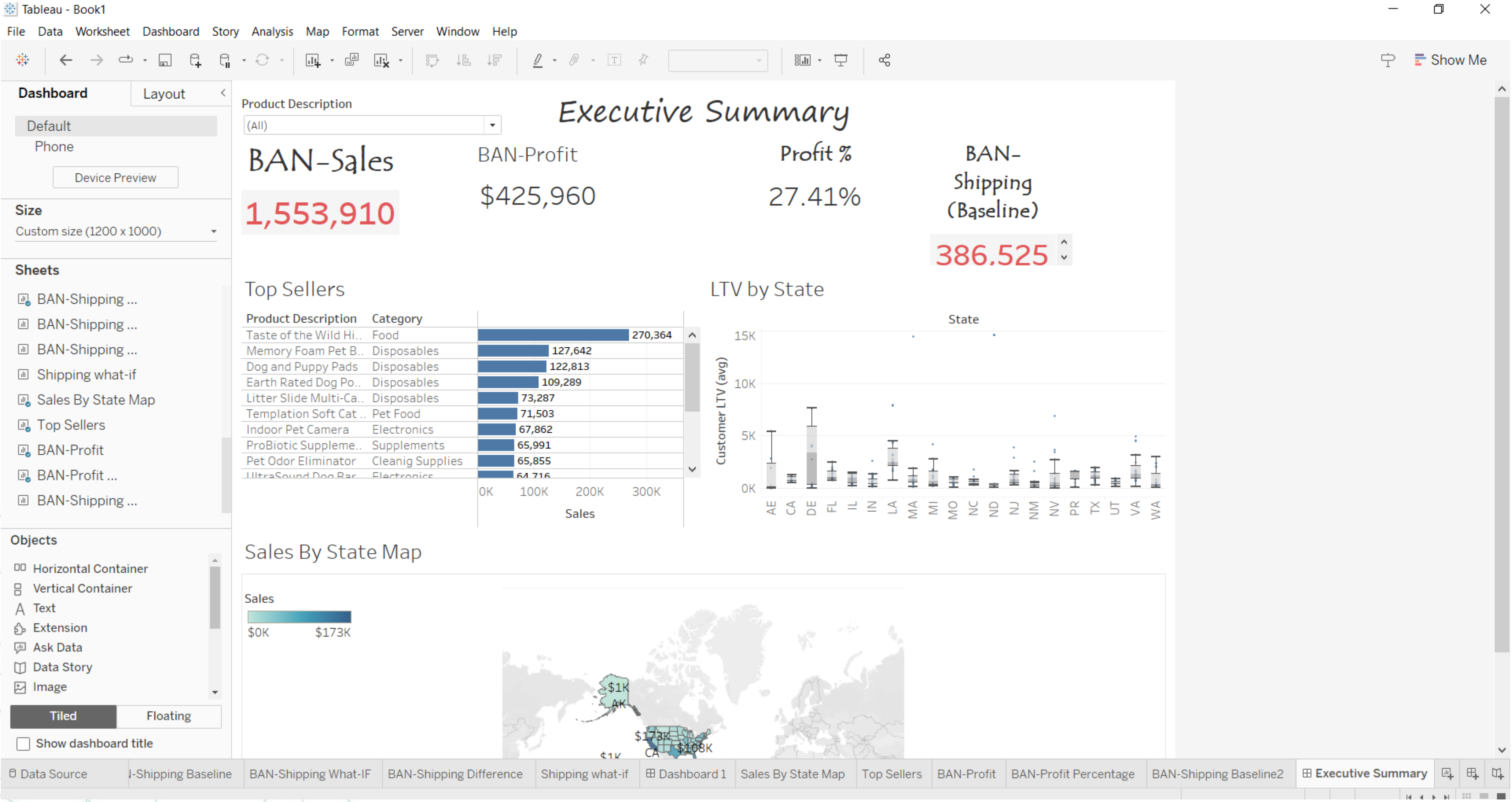


Step 46– Data Visualization & Storytelling

Align Your Dashboard

What is the profit % margin for Taste of the Wild High Prairie dog food in the state of California (code: CA)?

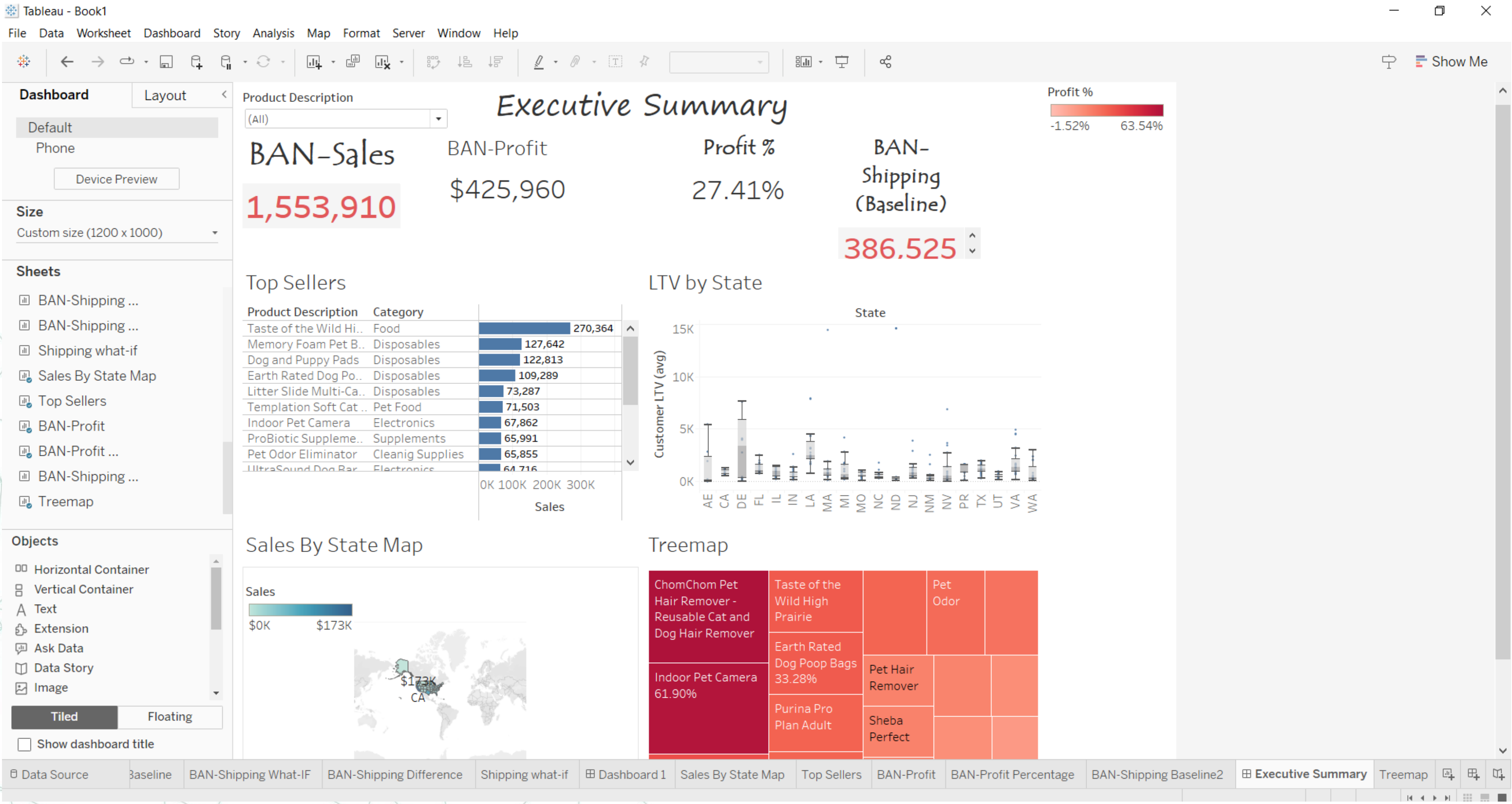
IF 26.26% proceed



Step 47– Data Visualization & Storytelling

- Profit percentage treemap
- The management has an enhancement request to the executive summary dashboard.
- They want you to add a breakdown of profit margin by product. You will create a new treemap, add it to the dashboard, and enable the action filter on the tree map.
- **Create a new sheet "Treemap"** of **Product Description** and size it by **Profit %**. Show **Product Description** and **Profit %** as text labels.
- Add the Treemap sheet to the Executive Summary you built in the last exercise.
- Set this sheet as an action filter on the dashboard.

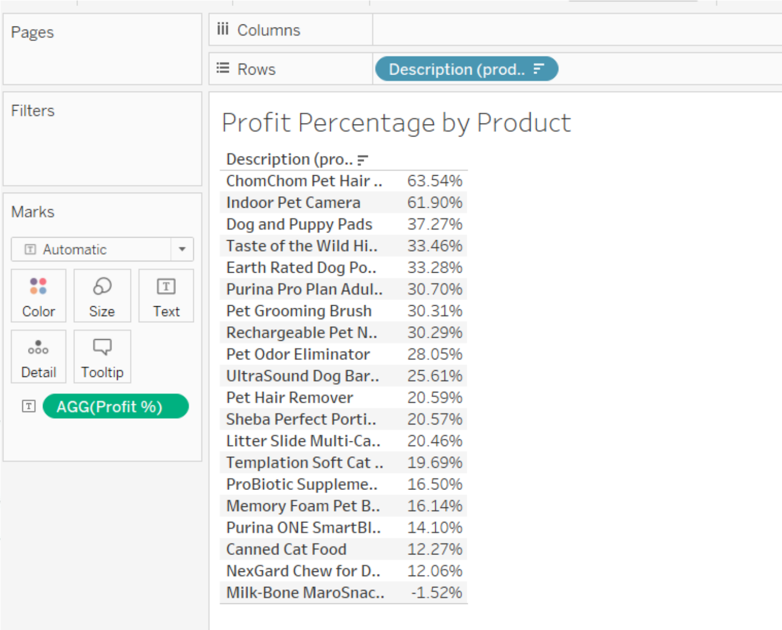
Step 47– Data Visualization & Storytelling



Step 48– Data Visualization & Storytelling

- Profit percentage
- Let us create a simple text table of profit margin by the products and then answer a question about it.

- Create a new sheet, "Profit Percentage by Product", and add a text table using Profit % and Product Description.
- Sort descending on Profit %.



Description (pro..	
ChomChom Pet Hair ..	63.54%
Indoor Pet Camera	61.90%
Dog and Puppy Pads	37.27%
Taste of the Wild Hi..	33.46%
Earth Rated Dog Po..	33.28%
Purina Pro Plan Adul..	30.70%
Pet Grooming Brush	30.31%
Rechargeable Pet N..	30.29%
Pet Odor Eliminator	28.05%
UltraSound Dog Bar..	25.61%
Pet Hair Remover	20.59%
Sheba Perfect Porti..	20.57%
Litter Slide Multi-Ca..	20.46%
Templation Soft Cat ..	19.69%
ProBiotic Suppleme..	16.50%
Memory Foam Pet B..	16.14%
Purina ONE SmartBl..	14.10%
Canned Cat Food	12.27%
NexGard Chew for D..	12.06%
Milk-Bone MaroSnac..	-1.52%

Step 49– Data Visualization & Storytelling

- Growth opportunities dashboard

  You are now reaching the finish line. Please wear your analyst hat and provide recommendations on which products should be recommended to customers on the checkout page for cross-sell promotions.

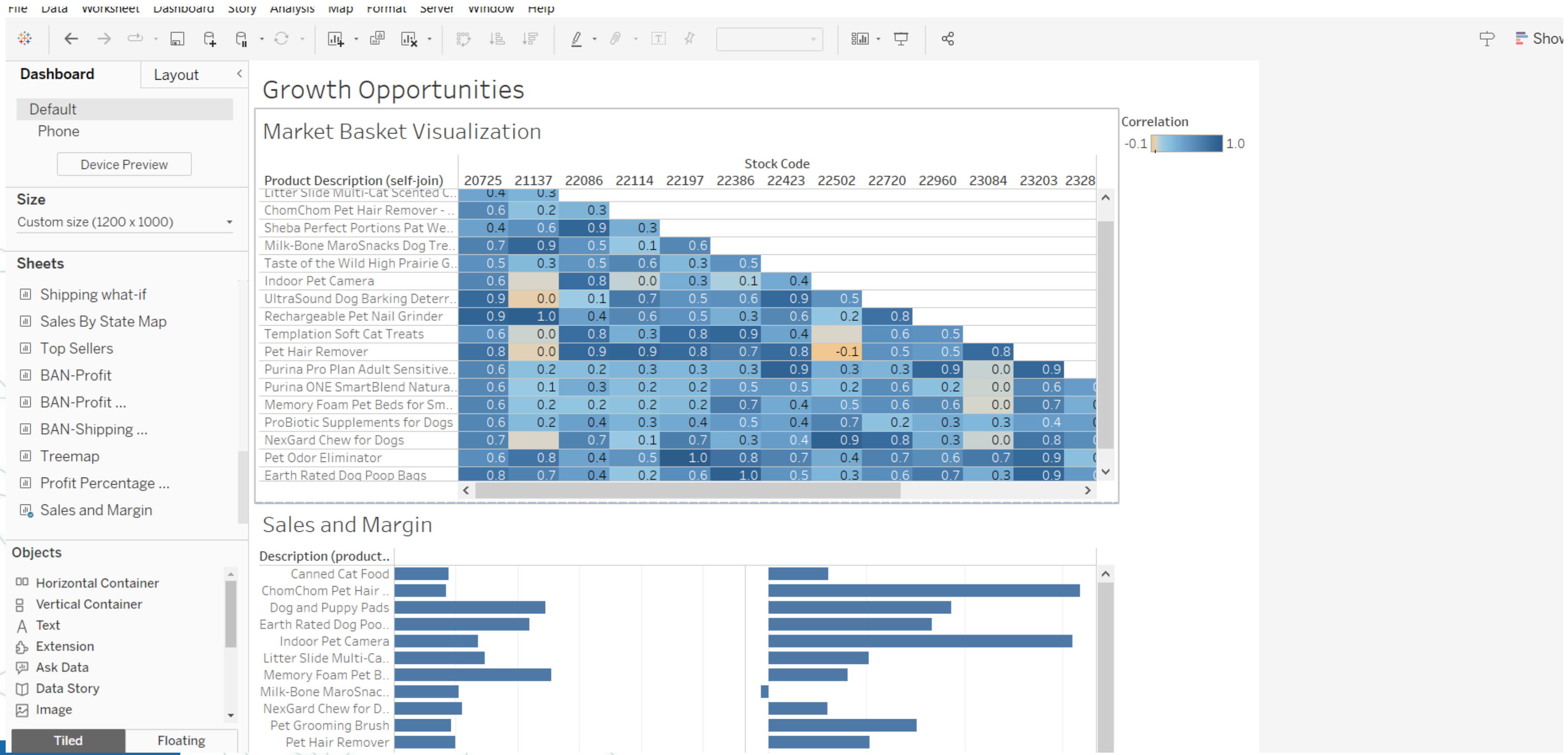
- You will use a combination of the market basket correlation matrix built before and a sales chart you will build to provide your suggestions.

- Using the Product Sales data source, create a new sheet, "Sales and Margin", and add a horizontal bar chart displaying Production Description by Sales and Profit % as two side-by-side horizontal bars.

- Create a new dashboard, "Growth Opportunities", and add Market Basket Visualization.
- Adjust the tooltip to display both product descriptions on the correlation matrix.
- Set a custom dashboard size to at least "1200×1000" or set the size as per your preference.
- Add Sales and Margin to the dashboard, set subtitles, and adjust layout and formatting.

Step 50 – Data Visualization & Storytelling

- Analyze the correlation matrix data to understand which products are frequently bought together.
- Combine this insight with sales amount to make one cross-sell recommendation for the management.
- UltraSound Dog Barking Deterrent has a correlation of 0.5 with two products:Sheba Perfect Portions Cat FoodIndoor Pet CameraWhich product would you recommend if you had to pick one?



Step 51 – Data Visualization & Storytelling

Shipping Costs Dashboard

- You want to provide comprehensive suggestions for savings on shipping costs.
- You will visualize shipping costs breakdown by product and geography.
- You will also provide your recommendations on quantity upsell strategies.

Create Two New Sheets for Shipping Cost Analysis

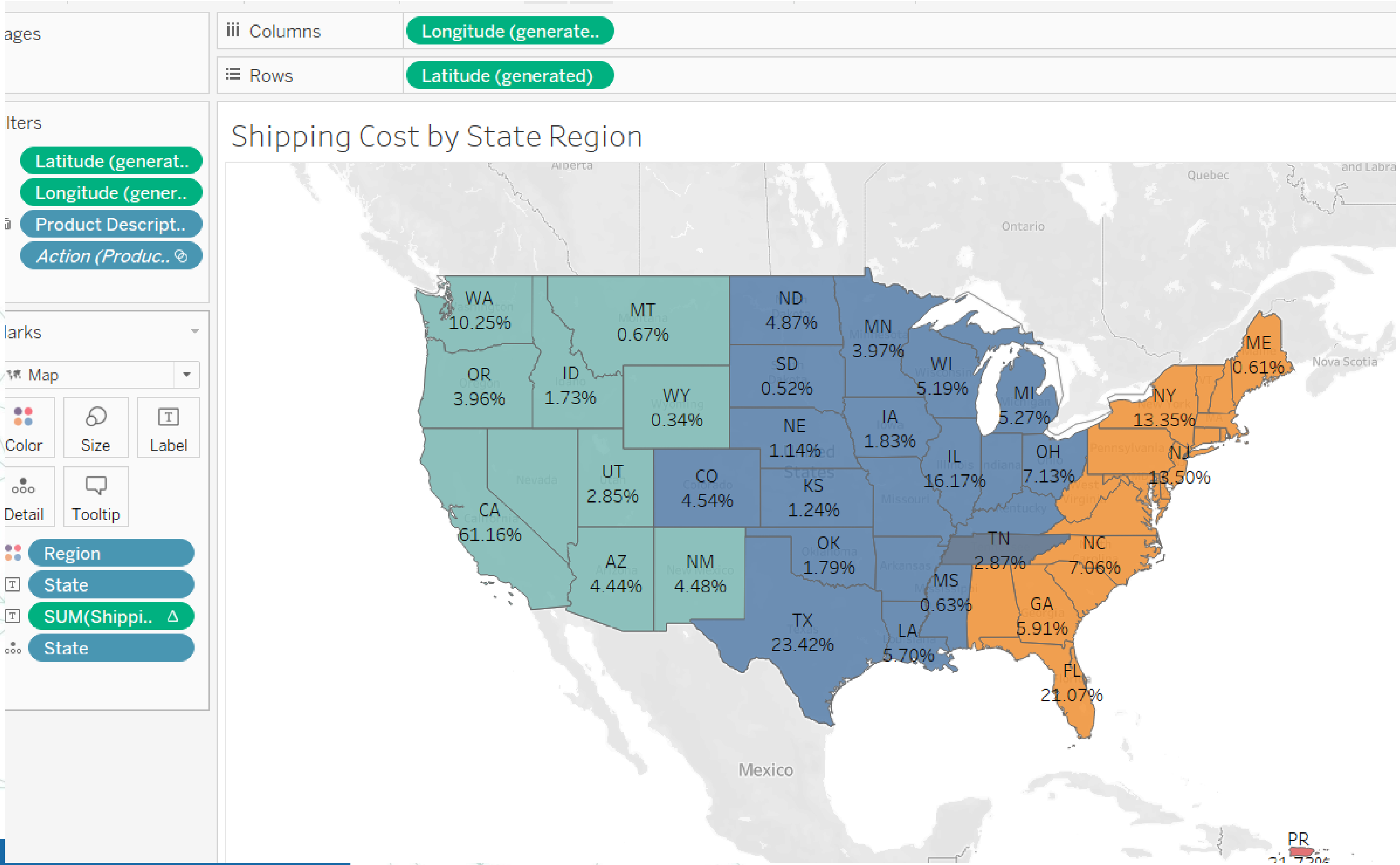
1.Duplicate "Treemap"

1. Rename it as **"Shipping Cost by Product"**.
2. Replace **Profit %** with **Shipping (Baseline)**.

2.Duplicate "Sales by State Map"

1. Rename it as **"Shipping Cost by State Region"**.
2. Replace **Sales** with **Shipping (Baseline)**.
3. Color by **Region**.
4. Add **Percent of Total** as a label.

Step 51 – Data Visualization & Storytelling



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Step 52 – Data Visualization & Storytelling

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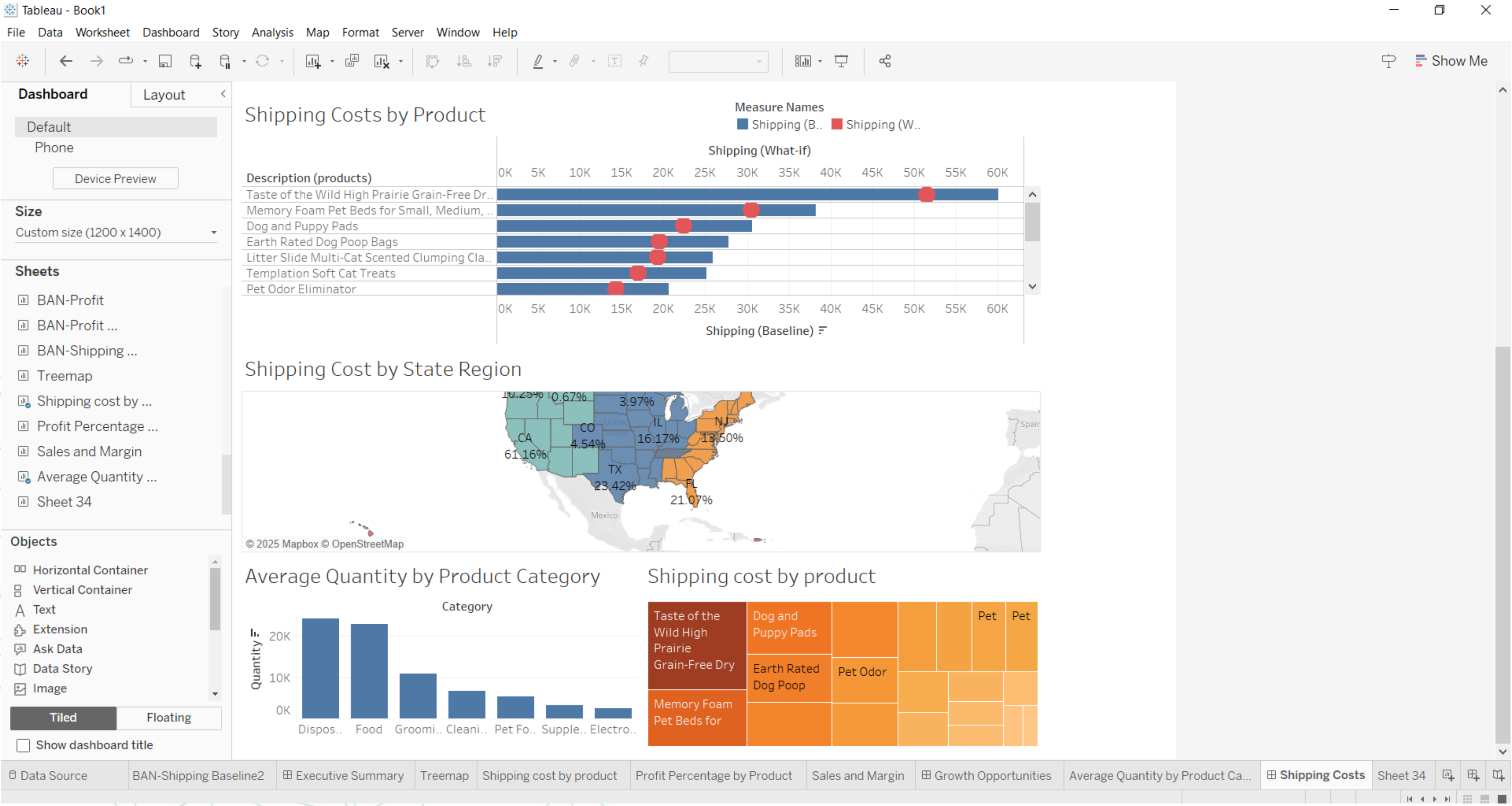
- **Create a new sheet**
  - Name it "Average Quantity by Product Category".
  - Add a bar chart showing the average Quantity against Category.
- **Duplicate "What-if Analysis" Dashboard**
  - Rename it as "Shipping Costs".
  - Move the duplicated dashboard to the rightmost position and increase its height to 1400.
  - Add the three new sheets created “**Shipping Cost by State Region” , “Shipping Cost by Product” and “Average Quantity by Product Category”** to the dashboard.
  - Change the dashboard title to "Shipping Cost Savings and Upsell Opportunities".

Step 52 – Data Visualization & Storytelling



Step 53 – Present Your Story

- you will build a data visualization story, analyze the data, and make concrete business strategy recommendations. Your recommendations should address questions about which products to market, how to save on shipping costs, and where additional investments should be made.
- **Create a new story "Pet Supply Analysis"**
- **Add Your Dashboards**

Step 53 – Present Your Story

On the "Executive Summary", what is the sales amount for the second best seller Memory Foam Pet bed?

On the "Executive Summary", what is the median lifetime value of customers in the state of Texas?

On the "Growth Oppotunities" dashboard, what is the correlation between UltraSound Dog Barking Deterrent with Taste of the Wild High Prairie Food?

Between the "East" and "Central" regions, which region should be prioritized for the opening of a new warehouse location?

Let's practice

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Any Questions ?

Thank You